

BALANCING PRIVACY AND UTILITY IN CORRELATED DATA: A STUDY OF BAYESIAN DIFFERENTIAL PRIVACY

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ABSTRACT. Privacy risks in differentially private (DP) systems increase significantly when data is correlated, as standard DP metrics often underestimate the resulting privacy leakage, leaving sensitive information vulnerable. Given the ubiquity of dependencies in real-world databases, this oversight poses a critical challenge for privacy protections. Bayesian differential privacy (BDP) extends DP to account for these correlations, yet current BDP mechanisms indicate notable utility loss, limiting its adoption.

In this work, we address whether BDP can be realistically implemented in common data structures without sacrificing utility—a key factor for its applicability. By analyzing arbitrary and structured correlation models, including Gaussian multivariate distributions and Markov chains, we derive practical utility guarantees for BDP. Our contributions include theoretical links between DP and BDP and a novel methodology for adapting DP mechanisms to meet the BDP requirements. Through evaluations on real-world databases, we demonstrate that our novel theorems enable the design of BDP mechanisms that maintain competitive utility, paving the way for practical privacy-preserving data practices in correlated settings.

1. INTRODUCTION

Differential privacy (DP) [15] has become the leading framework for preserving privacy in data analysis, providing formal guarantees that protect individuals’ sensitive information. However, its protection guarantees are limited to statistically independent data records, i.e., DP mechanisms can leak private information when the underlying data is correlated. The limitations of DP for protecting correlated data have been theoretically exposed [22, 27, 30, 39] and empirically confirmed with attacks on real databases [21]. This is a significant issue, as correlations among data records are common in real-world databases, such as those induced by friendships in social networks [26] or genetic similarities among family members [1].

Bayesian differential privacy (BDP) [41] extends DP to account for data correlations. DP assumes the attacker’s prior knowledge includes all records except the target record. Thus, DP leakage estimation corresponds to the maximum ratio between output distributions for databases differing by a single record. In contrast, BDP considers all possible priors, including cases where unknown records are correlated. Consequently, BDP ensures that the ratio between output distributions remains limited even when the set of unknown correlated records is a superset of the target record. When records are statistically independent, DP and BDP yield the same leakage measure. However, in the presence of correlations, BDP quantifies the worst-case increase in distinguishability by integrating the mechanism’s output distribution with the data distribution via Bayes’ rule. This accounts for scenarios where correlations enhance the adversary’s ability to infer the target record more precisely than DP leakage predicts.

While BDP provides a robust framework for assessing privacy leakage under data dependencies, its practical applicability remains uncertain. The few mechanisms that satisfy this notion [7, 41] are limited to specific correlation models, such as Gaussian Markov random fields—a subclass of multivariate Gaussian distributions forming a Markov random field where missing edges correspond to zeros in the inverse covariance matrix [34]—and binary-state Markov chains. Given the scarcity of mechanisms and their restriction to specific distributions, it remains unclear

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whether BDP can serve as a usable privacy notion or if its applicability is confined to particular cases. Moreover, the reported utility is highly conservative, as noise addition scales linearly with the number of records in the database. The only proposed way to mitigate this utility loss is to weaken privacy by relaxing BDP and incorporating assumptions about the adversary’s prior knowledge [41].

In summary, original DP privacy leakage estimation does not provide sufficient protection under data dependencies, and there is a need for improved utility with the robust BDP framework. Motivated by this issue, this paper examines BDP’s utility from both theoretical and practical perspectives, analyzing its limitations and proposing new strategies to reduce utility loss while maintaining BDP privacy guarantees.

To address the aforementioned issues, we present theoretical bounds on the accuracy of BDP mechanisms and derive specific utility guarantees when certain correlation models are assumed. To formally analyze utility, we use the standard utility metric for DP mechanisms, (α, β) -accuracy [15, 40], due to its mathematical formalism and broad applicability. For the experimental results, we focus on two specific, albeit common, tasks: counting queries and sum queries [15].

We prove that, without any assumption about the data correlation model, it is impossible to build a BDP mechanism that provides reasonable accuracy and theoretically sound privacy at the same time. This impossibility result holds for any BDP mechanism and query function, indicating that it is not sensible to apply BDP without further restrictions. Thus, the rest of our work examines whether targeting specific correlation models can improve utility.

Particularly, we analyze the impact of limiting the amount of correlated records, and we investigate the applicability of BDP to both discrete and continuous correlation models. For the discrete case, we analyze data following a Markov chain and, for continuous data, we analyze multivariate Gaussian correlation. We focus on these two particular correlation models following previous work in BDP [25, 41] and due to their relevance in many real-world applications such as medical [5], location [17], or activity data [13].

For each correlation model studied, we prove novel theorems that bound the BDP leakage of a DP mechanism. Notably, our BDP leakage bound for Gaussian multivariate models is tighter than that provided in [41], and our correlation model is broader. These privacy bounds provide a systematic way to build BDP mechanisms by adjusting the parameters of existing DP mechanisms. Using this approach, we propose novel BDP mechanisms based on Laplace noise. Furthermore, we calculate the accuracy of our BDP mechanisms showing the improved accuracy compared to scenarios where protection is required against any correlation.

Finally, we provide insight into how our theoretical results apply in practice to real-world data containing Gaussian and Markov correlations. This allows us to confirm that our results enhance the utility of BDP mechanisms in actual applications.

In summary, this work makes the following main contributions:

- We prove a novel bound on the best accuracy achievable by any BDP mechanism against arbitrary correlation, showing the limitations of such an approach and emphasizing the need to assume a specific distribution to preserve utility.
- We prove the bound on the BDP leakage of a DP mechanism with a fixed number of arbitrarily correlated records. We call this the *general bound* and prove that it is tight for arbitrary correlations, even when the correlation strength is arbitrarily small.
- We derive a novel bound on the BDP leakage of DP mechanisms for a multivariate Gaussian correlation model, improving upon the general bound and the state of the art. This provides a systematic method for constructing BDP mechanisms that account for Gaussian correlations, resulting in improved accuracy.
- We derive a bound on the BDP leakage of DP mechanisms when data follows a Markov chain, improving on the general bound when the transition probabilities of the chain are sufficiently similar. This result offers a systematic approach to building BDP mechanisms that enhance accuracy under Markovian dependencies.

The paper is organized as follows: In Sections 2 and 3, we review relevant prior work and provide the necessary preliminaries. We then present our analysis of arbitrary correlation in Section 4, discussing the relationship between BDP, free-lunch, and the accuracy impossibility result. In Section 5, we explore the accuracy improvement achieved by limiting the number of correlated records. In Section 6, we analyze the impact of Gaussian correlation on BDP and provide our improved bound in Theorem 6.9. In Section 7, we present analogous results for the Markov scenario. Finally, we discuss our empirical study in Section 8, demonstrating the practical relevance of our theoretical results, and conclude with a brief summary in Section 9.

The code used for our experiments is available in our Github repository (<https://github.com/lange-martin/privacy-utility-bdp>).

2. RELATED WORK

Kifer and Machanavajjhala [22] first noted that real analyses frequently contain correlated records, demonstrating that DP leakage fails to account for the risk of membership inference under such circumstances. They introduced the *free-lunch* privacy notion, which aims to protect against arbitrary data distributions. While they provided intuition about the limitations of this approach, they did not formalize its impact on utility.

Other advanced privacy notions were designed to explicitly address data correlations [11, 20, 25, 27]. However, most impose specific structural assumptions on the correlation model such as dependent differential privacy [27] and blowfish privacy [20]. BDP [41] is a noteworthy exception to this rule. Additionally, BDP serves as a foundation for various privacy notions such as prior DP [25] and correlated DP for location data [11].

BDP presents a formalism that better captures the effects of correlation compared to standard differential privacy (DP). It maintains equivalence to DP for independent data. The authors pioneered the approach of calibrating noise in the Laplace mechanism based on a correlation model to achieve BDP protection and established a preliminary relationship between DP and BDP for Laplace mechanisms applied to Gaussian Markov random fields. While these contributions are foundational, the approach is limited to the case of Gaussian Markov random fields. Moreover, the BDP bound implies a linearly increasing privacy loss with respect to the dataset size, which intuitively leads to poor utility. Although the authors suggest limiting the adversary’s knowledge to mitigate utility loss, this weakens the adversary model and reduces privacy assurances. Finally, they leave empirical validation unaddressed.

Chakrabarti et al. [7] extend this with a novel BDP mechanism for Markov chains. Their approach, however, is only applicable to *lazy* Markov chains, where remaining in the current state is more likely. In addition, it assumes binary states and positive linear correlation, limiting applicability. Moreover, while they assess the relationship between DP and BDP within their specific mechanism, they do not establish any general connection between DP and BDP for Markov chains.

In conclusion, while previous work highlights the limitations of DP protection and the need for BDP as a privacy standard, the challenge of improving utility with BDP protection remains unresolved, and the relationship between DP and BDP is not fully understood. Also, the effect of free-lunch privacy on the accuracy of any mechanism, and its connection to BDP so far are not investigated. We hence address these points in this work.

3. BACKGROUND

In this section, we present the fundamental definitions and notation necessary for the proper understanding of this work. The key notation used throughout this paper is shown in Table 1.

3.1. Differential Privacy and Metric Privacy. In the original formulation of DP [15], the database is assumed to be comprised of a finite number n of rows, $D = (x_1, \dots, x_n) \in \mathcal{X}^n$, where the intuition is that each row contains data related to an individual, drawn from a universe of data records $\mathcal{X}$. We use $[n]$ to denote the set of indices $[n] := \{1, \dots, n\}$ in the database. The formalization ϵ -DP introduced by Dwork and Roth reads as follows:

Notation	Description
$\mathcal{X}$	Domain of a single record $x \in \mathcal{X}$.
$\mathcal{M} : \mathcal{X}^n \rightarrow \mathcal{Y}$	Randomized mechanism with input from domain $\mathcal{X}^n$ and output in codomain $\mathcal{Y}$.
$\mathbf{X} = (X_1, \dots, X_n)$	Random vector representing the input of $\mathcal{M}$.
Y	Random variable representing output of $\mathcal{M}$.
$[n]$	Set $\{1, \dots, n\}$ for $n \in \mathbb{N}$.
$\mathbf{X}_K = (X_{i_1}, \dots, X_{i_k})$	Random vector of a subset $K = \{i_1, \dots, i_k\} \subseteq [n]$ of the random variables $X_1, \dots, X_n$.
$\mathbf{x}_K = (x_{i_1}, \dots, x_{i_k}) \in \mathcal{X}^k$	Database with k records.

TABLE 1. Notation

Definition 3.1 (Differential Privacy [15]). A randomized mechanism $\mathcal{M} : \mathcal{X}^n \rightarrow \mathcal{Y}$ is called ε -differentially private, if for all measurable sets $S \subseteq \mathcal{Y}$ and for all pairs of input databases $D, D' \in \mathcal{X}^n$ with $d_H(D, D') = 1$ we have

$$\Pr_{\mathcal{M}}[Y \in S \mid D] \leq e^\varepsilon \Pr_{\mathcal{M}}[Y \in S \mid D'],$$

where $d_H(D, D')$ denotes the Hamming distance.

The output of $\mathcal{M}$ is represented by the random variable Y , which depends on the input data D . The privacy leakage ε governs the privacy-utility trade-off: a smaller ε means that the output distributions for neighboring inputs are “closer together”. It results in higher privacy for the participants with an opposing effect on utility (See Proposition 3.5).

All $D, D' \in \mathcal{X}^n$ such that $d_H(D, D') = 1$ are called *bounded-neighboring databases* [22]. This neighborhood definition specifies which information can change, while ensuring that the output probabilities remain similar up to e^ε . In the bounded case, we assume the attacker knows all data records except for the target index $i \in [n]$, for which all pairs of possibilities x_i, x'_i must be indistinguishable. Following this intuition, the DP definition becomes equivalent to stating that for all measurable sets $S \subseteq \mathcal{Y}$, any target index $i \in [n]$, any target values $x_i, x'_i \in \mathcal{X}$, and any remaining values $\mathbf{x} \in \mathcal{X}^{n-1}$, we have

$$\Pr[Y \in S \mid \mathbf{X}_{-i} = \mathbf{x}, X_i = x_i] \leq e^\varepsilon \Pr[Y \in S \mid \mathbf{X}_{-i} = \mathbf{x}, X_i = x'_i].$$

Here, $\mathbf{X} = (X_1, \dots, X_n)$ is a random vector and D is drawn from the distribution of $\mathbf{X}$. For a set of indices $K = \{i_1, \dots, i_k\} \subseteq [n]$, we define the random vector $\mathbf{X}_K \in \mathcal{X}^k$ as $\mathbf{X}_K := (X_{i_1}, \dots, X_{i_k})$, particularly $\mathbf{X}_{-i}$ denotes $\mathbf{X}_K$ with $K = [n] \setminus \{i\}$. While this definition of DP is mathematically equivalent, it gives a better intuition of the interpretation of DP in comparison to BDP.

We focus on a bounded neighborhood due to its broad applicability and its close relation to BDP. However, other neighboring definitions exist [12]. For instance, in streaming data applications it is common to use *event-level DP* [15]: While each stream belongs to an individual, two streams are neighbors if they differ in one single time step value. We will see an example of application of this neighborhood in Section 8. The change of neighborhood allows to encode protection against different privacy threats [8, 12]. To obtain a general framework suitable to model a large variety of privacy problems, Chatzikokolakis et al. [8] introduce *metric privacy* as a generalization of DP that encapsulates the neighborhood notion and privacy leakage ε into a single parameter d , which determines the level of indistinguishability between databases:

Definition 3.2 (Metric Privacy [8]). Given $d : \mathcal{X}^{2n} \rightarrow \mathbb{R}$ a pseudometric, a randomized mechanism $\mathcal{M} : \mathcal{X}^n \rightarrow \mathcal{Y}$ is called d -private if for all databases $D, D' \in \mathcal{X}^n$ and all measurable sets $S \subseteq \mathcal{Y}$ we have

$$\Pr_{\mathcal{M}}[Y \in S \mid \mathbf{X} = D] \leq e^{d(D, D')} \Pr_{\mathcal{M}}[Y \in S \mid \mathbf{X} = D'].$$

This definition makes it challenging for an adversary to distinguish between databases D and D' that are “close” according to the metric d . However, if the two databases are significantly different, the output distributions can differ more, making it easier for the adversary to distinguish them. Note that d -privacy is equivalent to DP when considering the Hamming distance scaled by ε .

There exist numerous mechanisms in the literature that achieve DP while giving a good approximation of the target global query answer $f(D)$. Hereafter, we focus on the Laplace mechanism [15], one of the earliest and most common methods proven to satisfy ε -DP. This mechanism relies on the sensitivity of the function that accounts for the biggest impact that the change of a data record can have on the target global query.

Definition 3.3 (Laplace Mechanism [15]). Let $f : \mathcal{X}^n \rightarrow \mathbb{R}^k$ be a function and its *sensitivity* defined as

$$\Delta f := \sup_{d_H(D,D')=1} \|f(D) - f(D')\|_1.$$

Given that sensitivity $\Delta f < \infty$ and $\varepsilon > 0$, the Laplace mechanism is defined for all $D \in \mathcal{X}^n$ as:

$$\mathcal{M}_{\varepsilon,f}(D) = f(D) + (Z_1, \dots, Z_k)$$

where Z_i are i.i.d. random variables that follow the Laplace distribution centered at 0 and with scale $\frac{\Delta f}{\varepsilon}$.

While $\mathcal{M}_{\varepsilon,f}$ provides ε -DP, adding noise to the output of a function f undoubtedly has an impact on utility. A well-established metric for quantifying the utility of a private mechanism is the (α, β) -accuracy [4, 27]. It provides a measure of how well the mechanism approximates a true statistic or function while considering the inherent randomness introduced by the mechanism. More precisely, it refers to the utility guarantee that with probability at least $1 - \beta$, the mechanism’s output is within an interval of radius α centered on the true value.

Definition 3.4 ((α, β) -Accuracy [4]). A randomized mechanism $\mathcal{M}$ is (α, β) -accurate with respect to function f if for all databases $D \in \mathcal{X}^n$ we have

$$\Pr[|\mathcal{M}(D) - f(D)| \geq \alpha] \leq \beta.$$

In words, a randomized mechanism $\mathcal{M}$ is (α, β) -accurate if an error of magnitude α has a probability of at most β . Thus, the smaller α and/or β , the better the accuracy of mechanism $\mathcal{M}$. Particularly, α represents the error tolerance. It defines how close the output of the differentially private mechanism is expected to be to the true value. While β represents the failure probability. It defines the likelihood that the mechanism’s output will not fall within the specified error margin α . Practically speaking, an (α, β) -accurate mechanism ensures that for at least $1 - \beta$ of the times the mechanism is used, its output will be within the error bound α of the true result.

We can calculate the (α, β) -accuracy of a mechanism that utilizes the Laplace mechanism as follows:

Proposition 3.5 ([15]). Let $\mathcal{M}_{\varepsilon,f}$ be the Laplace mechanism. Let $\beta \in (0, 1]$ be a probability. Then $\mathcal{M}_{\varepsilon,f}$ is (α, β) -accurate with respect to f with

$$\alpha = \ln\left(\frac{1}{\beta}\right) \cdot \frac{\Delta f}{\varepsilon}.$$

This accuracy result for the Laplace mechanism is tight [15].

3.2. Bayesian Differential Privacy. BDP [41] extends the privacy protection goal of DP to correlated data. It assumes the adversaries are uncertain between two possible records x_i, x'_i to be in the database, analogously as DP. However, it eliminates the notion of neighboring databases in order to consider different possible adversaries with different background knowledge. Formally, the adversary (K, i) is targeting the record at position i and already knows the values of the sub vector $\mathbf{x}_K = (x_{i_1}, \dots, x_{i_k})$ on the database. Then, for each possible adversary we can define its Bayesian leakage as follows:

Definition 3.6 (Adversary-specific BDPL [41]). Given $\mathcal{M} : \mathcal{X}^n \rightarrow \mathcal{Y}$ a randomized mechanism, $\mathbf{X}$ the input random vector following the distribution π , the targeted record index $i \in [n]$, and the known record indices $K \subseteq [n] \setminus \{i\}$, the *adversary-specific Bayesian differential privacy leakage* is

$$\text{BDPL}_{(K,i)} = \sup_{x_i, x'_i, \mathbf{x}_K, S} \ln \frac{\Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x_i]}{\Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x'_i]},$$

where the supremum is taken over all the possible target values $x_i, x'_i \in \mathcal{X}$, all the possible known vector values $\mathbf{x}_K \in \mathcal{X}^K$ and all the measurable sets $S \subseteq \mathcal{Y}$.

When computing the adversary-specific BDPL, the correlation between the unknown and known records modifies the final leakage since given the unknown remaining indexes U , we have

$$\Pr[Y \in S \mid \mathbf{x}_K, x_i] = \sum_{\mathbf{x}_U \in \mathcal{X}^u} \Pr[Y \in S \mid \mathbf{x}_K, x_i, \mathbf{x}_U] \Pr[\mathbf{x}_U \mid \mathbf{x}_K, x_i],$$

where $u = |U| = n - k - 1$. The sum must be substituted by an integral in the continuous case.

While the adversary-specific BDPL only accounts for a particular case, we aim to protect against any possible adversary. Therefore, to compute the worst-case leakage we take the supremum over all possible adversaries' leakages:

Definition 3.7 (Bayesian DP [41]). A mechanism $\mathcal{M}$ is an ε -*Bayesian differentially private* mechanism if

$$\text{BDPL}(\mathcal{M}) = \sup_{K,i} \text{BDPL}_{(K,i)}(\mathcal{M}) \leq \varepsilon,$$

where the supremum is taken over all the possible set of indexes $i \in [n]$ and $K \subseteq [n] \setminus \{i\}$. $\text{BDPL}(\mathcal{M})$ is called *Bayesian differential privacy leakage*.

The BDPL has a similar role to the privacy leakage ε in DP: It measures the extent of a possible privacy violation by comparing the difference in the output probabilities of mechanism $\mathcal{M}$. A lower BDPL corresponds to higher privacy because any adversary will be less likely to differentiate between any two target values $x_i, x'_i \in \mathcal{X}$. Particularly, if X_i, X_j are mutually independent for all $i \neq j \in [n]$ then ε -DP and ε -BDP are equivalent [41].

While we have results on the accuracy loss associated with using DP mechanisms (Proposition 3.5), the impact of BDP protection on utility remains unclear. The primary goal of this work is to address this question by analyzing various correlation scenarios. These results are presented in Sections 4 to 7.

4. ARBITRARY CORRELATION

It is often unclear whether a given statistical model is an appropriate approximation for the distribution and correlation of a database [37]. In such cases, it would be beneficial to mitigate the impact of any potential correlation without making assumptions about the data distribution. That means that our protection mechanism must be BDP with respect to any correlation distribution π . We call this strategy *protection with respect to arbitrary correlation*. Therefore, we investigate the best (α, β) -accuracy that *any* ε -BDP mechanism that protects against arbitrary correlation can achieve. Unfortunately, our results show that BDP mechanisms cannot guarantee privacy with respect to arbitrary correlation and provide utility at the same time by reducing the case to *free-lunch privacy* [22].

The fundamental goal of DP is to compute a global statistic $f(D)$ while keeping individual records $x \in D$ private. Free-lunch privacy is a special case of d -privacy with $d(D, D') = \varepsilon$ for all possible pairs of $D, D' \in \mathcal{X}^n$. This ensures ε -indistinguishability between *any* two databases, providing protection against *any* adversary, including those with knowledge of correlations. However, this comes at the cost of utility: an ε -free-lunch private mechanism obscures almost all information due to excessive noise [22]. A Free-lunch private mechanism not only conceals individual information but necessarily also makes the global statistic $f(D)$ indistinguishable from any other possible value $f(D')$. In other words, enforcing indistinguishability between all databases leads to indistinguishability of every information we aim to extract [14]. As a

result, the privacy-utility trade-off becomes as unfavorable as possible: we lose utility in direct proportion to the privacy gained, ultimately rendering the mechanism impractical.

We formalize this intuition in terms of the (α, β) -accuracy, providing a lower bound for the error tolerance α of any ε -free-lunch private mechanism:

Theorem 4.1. *Let $\mathcal{M} : \mathcal{X}^n \rightarrow \mathbb{R}$ be an ε -free-lunch private mechanism. Let $0 \leq \beta < \frac{1}{e^\varepsilon + 1}$ be a real number and let $f : \mathcal{X}^n \rightarrow \mathbb{R}$ be a deterministic function. If mechanism $\mathcal{M}$ is (α, β) -accurate w.r.t. f for an error $\alpha \geq 0$, then*

$$\alpha > \frac{1}{2} \max_{\tilde{D}, \tilde{D}'} |f(\tilde{D}) - f(\tilde{D}')|.$$

Proof. We need to prove that given $\beta < \frac{1}{e^\varepsilon + 1}$, we obtain $\alpha > \frac{1}{2} |f(D) - f(D')|$ for each pair of databases $D, D' \in \mathcal{X}^n$, which then implies $\alpha > \frac{1}{2} \max_{D, D' \in \mathcal{X}^n} |f(D) - f(D')|$. We prove this by *reductio ad absurdum*. We assume that $\mathcal{M}$ fulfills an (α, β) -accuracy respect to f with $\alpha \leq \frac{1}{2} |f(D) - f(D')|$ and $\beta < \frac{1}{e^\varepsilon + 1}$. Under (α, β) -accuracy assumption, we have

$$\Pr[|\mathcal{M}(D) - f(D)| \geq \alpha] \leq \beta \quad (4.1)$$

for all $D \in \mathcal{X}$, and in particular for D' , by Definition 3.4. However,

$$\begin{aligned} \Pr[|f(D') - \mathcal{M}(D')| \geq \alpha] &= \Pr[\mathcal{M}(D') \in \mathbb{R} \setminus (f(D') - \alpha, f(D') + \alpha)] \\ &\geq \Pr[\mathcal{M}(D') \in (f(D) - \alpha, f(D) + \alpha)] \\ &\geq e^{-\varepsilon} \Pr[\mathcal{M}(D) \in (f(D) - \alpha, f(D) + \alpha)] \\ &= e^{-\varepsilon} (1 - \Pr[\mathcal{M}(D) \in \mathbb{R} \setminus (f(D) - \alpha, f(D) + \alpha)]) \\ &= e^{-\varepsilon} (1 - \Pr[|f(D) - \mathcal{M}(D)| \geq \alpha]) \\ &\stackrel{(*)}{\geq} e^{-\varepsilon} (1 - \beta) \\ &= \frac{1}{e^\varepsilon + 1} > \beta \end{aligned}$$

Where $(*)$ follows from the starting assumption 4.1. Therefore, we obtain the contradiction with $\Pr[|\mathcal{M}(D') - f(D')| \geq \alpha] \leq \beta$. $\square$

Specifically, the result from Theorem 4.1 indicates that the probability of an error greater than $\alpha = \frac{1}{2} \max_{\tilde{D}, \tilde{D}'} |f(\tilde{D}) - f(\tilde{D}')|$ is at least $\beta = \frac{1}{e^\varepsilon + 1}$. This means that for theoretically relevant privacy levels $\varepsilon \in (0, 4)$ [24], the only confidence interval where we can reliably estimate the actual value of our query function f , with standard confidence levels (e.g., between 90% and 99%), includes almost all possible query values. The result indicates low utility since such a broad interval provides little meaningful information. In practical terms, we can not differentiate even the most extreme differences in f 's values.

For example, our goal is to count the number of infected individuals without revealing the health status of any patient. To achieve this, we use a free-lunch algorithm to compute $f(D)$, which counts the number of infections in a database of n individuals. However, we are likely to get outputs that are completely uninformative – for instance, we might obtain a result of $\frac{n}{2}$, suggesting that half of the population is infected. In this case, with a confidence interval of 90%, we cannot tell whether there is no infection at all, or whether the entire population is infected.

We have formalized the utility limitations of free-lunch mechanisms. Consequently, if we want to obtain accurate results from a BDP algorithm, we must ensure that it is not free-lunch. Unfortunately, this is not possible, as we prove below that any BDP mechanism designed to protect against arbitrary correlations is inherently free-lunch:

Theorem 4.2. *Any ε -BDP mechanism $\mathcal{M}$ protecting against arbitrary correlation satisfies ε -free-lunch.*

Proof. We need to prove that for every pair of databases $D = (x_1, \dots, x_n)$ and $D' = (x'_1, \dots, x'_n)$, and for all measurable sets $S \subseteq \mathcal{R}$, the following inequality holds:

$$\Pr[\mathcal{M}(D) \in S] \leq e^\varepsilon \Pr[\mathcal{M}(D') \in S]. \quad (4.2)$$

Since $\mathcal{M}$ is ε -BDP for arbitrary correlation models, it is sufficient to show that, for each pair of databases, we can define a distribution π such that if $\mathcal{M}$ is ε -BDP with respect to π , then Equation (4.2) holds.

If $D = D'$, then $\Pr[\mathcal{M}(D) \in S] = \Pr[\mathcal{M}(D') \in S]$ and so the result is trivially satisfied for all $\varepsilon \geq 0$. Therefore, we consider $D \neq D'$. This means there exists at least one $i \in [n]$ such that $x_i \neq x'_i$. We denote $D_{-i} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ and $D'_{-i} = (x'_1, \dots, x'_{i-1}, x'_{i+1}, \dots, x'_n)$. We consider any distribution π verifying that

$$\begin{aligned}\Pr_{\pi}[\mathbf{X}_{-i} = D_{-i} \mid X_i = x_i] &= 1, \\ \Pr_{\pi}[\mathbf{X}_{-i} = D'_{-i} \mid X_i = x'_i] &= 1.\end{aligned}$$

Since $\mathcal{M}$ is an ε -BDP mechanism with respect to any arbitrary correlation, it is, in particular, ε -BDP with respect to π . This means that the adversary-specific BDPL for any adversary (K, j) is bounded by ε . In particular, it is bounded for the adversary that targets index i and does not know any records $K = \emptyset$. Then, for any measurable set $S \subseteq \mathbb{R}$ we have

$$\begin{aligned}e^{\varepsilon} &\geq \frac{\Pr[Y \in S \mid x_i]}{\Pr[Y \in S \mid x'_i]} = \frac{\sum_{\mathbf{x}_{-i} \in \mathcal{X}^{n-1}} \Pr[Y \in S \mid \mathbf{x}_{-i}, x_i] \Pr[\mathbf{x}_{-i} \mid x_i]}{\sum_{\mathbf{x}_{-i} \in \mathcal{X}^{n-1}} \Pr[Y \in S \mid \mathbf{x}_{-i}, x'_i] \Pr[\mathbf{x}_{-i} \mid x'_i]} \\ &= \frac{\Pr[Y \in S \mid D_{-i}, x_i]}{\Pr[Y \in S \mid D'_{-i}, x'_i]} = \frac{\Pr[\mathcal{M}(D) \in S]}{\Pr[\mathcal{M}(D') \in S]}.\end{aligned}$$

Since the same process can be done for each pair D, D' of databases, we obtain the result. $\square$

The intuition is that, without making any assumptions about the correlation strength or the number of correlated records, we can always consider a distribution where the change in one record is strongly correlated with changes in all the others. In this scenario, via Bayes' rule, we end up ensuring the protection of replacing an entire dataset with another, which is exactly free-lunch as shown in the previous proof.

As a direct consequence of Theorems 4.1 and 4.2 we obtain that, without further assumptions on the correlation model, the accuracy of BDP mechanism must also be poor:

Corollary 4.3. *Let $\mathcal{M} : \mathcal{X}^n \rightarrow \mathbb{R}$ be an ε -BDP mechanism protecting against arbitrary correlation. Let $0 \leq \beta < \frac{1}{e^{\varepsilon} + 1}$ be a real number and let $f : \mathcal{X}^n \rightarrow \mathbb{R}$ be a deterministic function. If mechanism $\mathcal{M}$ is (α, β) -accurate w.r.t. f for an error $\alpha \geq 0$, then*

$$\alpha > \frac{1}{2} \max_{\tilde{D}, \tilde{D}'} |f(\tilde{D}) - f(\tilde{D}')|.$$

Proof. Since $\mathcal{M}$ is ε -BDP protecting with respect to any correlation, applying Theorem 4.2, we conclude that $\mathcal{M}$ is ε -free-lunch. Therefore, the result follows directly from applying Theorem 4.1. $\square$

The results of this section indicate that it is not possible to obtain good utility guarantees for BDP mechanisms when we assume that the data correlation is completely unknown to the data curator. However, this assumption implies that the data curator knows less about the data than the adversary since the BDPL is calculated for adversaries with perfect correlation knowledge. Therefore, the following sections cover situations where specific models of the data distribution can be assumed by the data curator.

5. LIMITED NUMBER OF CORRELATED VARIABLES

In many scenarios it is reasonable to assume that only subsets of records within a database are correlated. For instance, in the context of genomic data, an individual's genome is strongly correlated with that of their relatives, but not necessarily with the entire population. While genomes share common features across humans, the correlations typically occur within families or specific populations rather than universally [1]. Therefore, in this section we assume that only m of n records in the database are correlated with each other. We formalize this as follows:

Definition 5.1. We say the random vector $\mathbf{X} = (X_1, \dots, X_n)$ has *at most* $m \leq n$ *correlated random variables* if there exist disjoint sets of indices $C_1, \dots, C_r$ that make up $[n] = \bigcup_{l=1}^r C_l$ so that each set C_l has maximum cardinality $m \geq |C_l|$ for any $l \in [r]$, and for any $l \in [r]$, the random variables $\{X_j \mid j \in C_l\}$ are independent of the remaining random variables $\{X_j \mid j \in [n] \setminus C_l\}$.

This definition considers multiple groups of up to m correlated records as long as they do not “overlap”, i.e., the records in one group are independent of the records in the rest of groups. Otherwise, we do not make any further assumptions about the distribution of the data. This allows us to find acceptable utility guarantees in Corollary 5.4 as long as m is sufficiently small.

5.1. Relationship between DP and BDP. First, we introduce and prove the general bound of the BDPL of an ε -DP mechanism stating that an ε -DP mechanism $\mathcal{M}$ that takes data drawn from a distribution with at most m correlated random variables, is at most $m\varepsilon$ -BDP. Multiplying the DP privacy leakage by the number of correlated records to determine the maximum privacy leakage against attacks exploiting correlations has been a common practice [9, 27]. However, to the best of our knowledge, this method has not previously been proven to fulfill the privacy notion BDP. Moreover, we prove that this bound is tight.

Theorem 5.2 (The General Bound). *Let $n \in \mathbb{N}$ be the size of the database. Let $\mathbf{X} = (X_1, \dots, X_n)$ be a random vector with at most $m \leq n$ correlated random variables that follows a distribution π . Then, any ε -DP mechanism $\mathcal{M}$ with input data drawn from distribution π is $m\varepsilon$ -BDP.*

Proof. Consider any adversary (K, i) with $i \in [n]$, $K \subseteq [n] \setminus \{i\}$ and $k = |K|$. Since $\{C_j\}_{j \in [r]}$ is a partition of $[n]$, there exists an $l \in [r]$ so that we have target index $i \in C_l$. Thus, C_l contains the index i and all indices of random variables potentially correlated with X_i . Let set $\tilde{C} := C_l \setminus K$ be the indices of random variables correlated with X_i that are not already included in K . Then, we first show that the adversary-specific BDPL can be upper bounded as follows:

$$\text{BDPL}_{(K,i)} = \sup_{S, \mathbf{x}_K, x_i, x'_i} \ln \frac{\Pr[Y \in S \mid \mathbf{x}_K, x_i]}{\Pr[Y \in S \mid \mathbf{x}_K, x'_i]} \leq \sup_{S, \mathbf{x}_K, \mathbf{x}_{\tilde{C}}, \mathbf{x}'_{\tilde{C}}} \ln \frac{\Pr[Y \in S \mid \mathbf{x}_K, \mathbf{x}_{\tilde{C}}]}{\Pr[Y \in S \mid \mathbf{x}_K, \mathbf{x}'_{\tilde{C}}]} \quad (5.1)$$

Assume that for all $\mathbf{x}_{\tilde{C}} \in \mathcal{X}^{|\tilde{C}|}$, we have

$$\Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x_i] > \Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, \mathbf{X}_{\tilde{C}} = \mathbf{x}_{\tilde{C}}]. \quad (5.2)$$

Now, we bring this to a contradiction, thereby proving the opposite of eq. (5.2). Let index set $\tilde{C}_{-i} := \tilde{C} \setminus \{i\}$ include all indices of $\tilde{C}$ except for i . Then, we have

$$\begin{aligned} & \Pr[Y \in S \mid \mathbf{x}_K, x_i] \\ &= \int_{\mathcal{X}^{|\tilde{C}_{-i}|}} \Pr[Y \in S \mid \mathbf{x}_K, x_i, \mathbf{x}_{\tilde{C}_{-i}}] p_{\mathbf{X}_{\tilde{C}_{-i}}}(\mathbf{x}_{\tilde{C}_{-i}} \mid \mathbf{x}_K, x_i) d\mathbf{x}_{\tilde{C}_{-i}} \end{aligned} \quad (5.3)$$

$$= \int_{\mathcal{X}^{|\tilde{C}_{-i}|}} \Pr[Y \in S \mid \mathbf{x}_K, (x_i, \mathbf{x}_{\tilde{C}_{-i}})] p_{\mathbf{X}_{\tilde{C}_{-i}}}(\mathbf{x}_{\tilde{C}_{-i}} \mid \mathbf{x}_K, x_i) d\mathbf{x}_{\tilde{C}_{-i}} \quad (5.4)$$

$$< \int_{\mathcal{X}^{|\tilde{C}_{-i}|}} \Pr[Y \in S \mid \mathbf{x}_K, x_i] p_{\mathbf{X}_{\tilde{C}_{-i}}}(\mathbf{x}_{\tilde{C}_{-i}} \mid \mathbf{x}_K, x_i) d\mathbf{x}_{\tilde{C}_{-i}} \quad (5.5)$$

$$= \Pr[Y \in S \mid \mathbf{x}_K, x_i] \int_{\mathcal{X}^{|\tilde{C}_{-i}|}} p_{\mathbf{X}_{\tilde{C}_{-i}}}(\mathbf{x}_{\tilde{C}_{-i}} \mid \mathbf{x}_K, x_i) d\mathbf{x}_{\tilde{C}_{-i}} \quad (5.6)$$

$$= \Pr[Y \in S \mid \mathbf{x}_K, x_i]. \quad (5.7)$$

where the random variable X_i is already included in the condition, so only indices $\tilde{C}_{-i}$ need to be added. In eq. (5.4), we combine the two conditions $X_i = x_i$ and $\mathbf{X}_{\tilde{C}_{-i}} = \mathbf{x}_{\tilde{C}_{-i}}$ into one condition $\mathbf{X}_{\tilde{C}} = (x_i, \mathbf{x}_{\tilde{C}_{-i}})$. Notice that this is the same condition, just stated differently. Then, we use eq. (5.2), which applies to all $\mathbf{x}_{\tilde{C}} \in \mathcal{X}^{|\tilde{C}|}$, in eq. (5.5). Now, the first probability can be pulled out of the integral in eq. (5.6) as it no longer depends on the value $\mathbf{x}_{\tilde{C}_{-i}}$. The final eq. (5.7) follows because a probability density integrated over its entire domain is always 1. Note that if the variables are discrete the integral must be changed by a sum and the result follows analogously.

We have shown that the initial probability is strictly smaller than itself—a contradiction. Thus, the opposite of our assumption in eq. (5.2) must be true and there must exist a vector $\mathbf{x}_{\tilde{C}}$ so that

$$\Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x_i] \leq \Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, \mathbf{X}_{\tilde{C}} = \mathbf{x}_{\tilde{C}}]. \quad (5.8)$$

Analogously, we can show that there exists a vector $\mathbf{x}'_{\tilde{C}} \in \mathcal{X}^{|\tilde{C}|}$ with the opposite property

$$\Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x'_i] \geq \Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, \mathbf{X}_{\tilde{C}} = \mathbf{x}_{\tilde{C}}]. \quad (5.9)$$

Thus, with eq. (5.8) and eq. (5.9) we have Equation (5.1). For any values of $\mathbf{x}_K$ and x_i, x'_i included in the left supremum, we can always find values $\mathbf{x}_{\tilde{C}}$ and $\mathbf{x}'_{\tilde{C}}$ so that the ratio becomes greater or equal, and these values $\mathbf{x}_{\tilde{C}}, \mathbf{x}'_{\tilde{C}}$ are included in the supremum on the right-hand side.

Now, we apply Equation (5.1) to prove the following statement. Let the set $U = [n] \setminus (K \cup \tilde{C})$, with $u = |U|$, include all remaining indices. Since by hypotheses $|\tilde{C}| \leq |C_l| \leq m$, for any known values $\mathbf{x}_K \in \mathcal{X}^k$, the correlated values $\mathbf{x}_{\tilde{C}} \in \mathcal{X}^{|\tilde{C}|}$ and $\mathbf{x}'_{\tilde{C}} \in \mathcal{X}^{|\tilde{C}|}$ we have

$$\begin{aligned} & \Pr_{\mathcal{M}}[Y \in S \mid \mathbf{x}_K, \mathbf{x}_{\tilde{C}}] \\ &= \int_{\mathcal{X}^u} \Pr_{\mathcal{M}}[Y \in S \mid \mathbf{x}_K, \mathbf{x}_{\tilde{C}}, \mathbf{x}_U] p_{\mathbf{X}_U}(\mathbf{x}_U \mid \mathbf{x}_K, \mathbf{x}_{\tilde{C}}) d\mathbf{x}_U \\ &\leq \int_{\mathcal{X}^u} e^{m\varepsilon} \Pr_{\mathcal{M}}[Y \in S \mid \mathbf{x}_K, \mathbf{x}'_{\tilde{C}}, \mathbf{x}_U] p_{\mathbf{X}_U}(\mathbf{x}_U \mid \mathbf{x}_K, \mathbf{x}_{\tilde{C}}) d\mathbf{x}_U \\ &= e^{m\varepsilon} \int_{\mathcal{X}^u} \Pr_{\mathcal{M}}[Y \in S \mid \mathbf{x}_K, \mathbf{x}'_{\tilde{C}}, \mathbf{x}_U] p_{\mathbf{X}_U}(\mathbf{x}_U \mid \mathbf{x}_K) d\mathbf{x}_U \\ &= e^{m\varepsilon} \int_{\mathcal{X}^u} \Pr[Y \in S \mid \mathbf{x}_K, \mathbf{x}'_{\tilde{C}}, \mathbf{x}_U] p_{\mathbf{X}_U}(\mathbf{x}_U \mid \mathbf{x}_K, \mathbf{x}'_{\tilde{C}}) d\mathbf{x}_U \\ &= e^{m\varepsilon} \Pr[Y \in S \mid \mathbf{x}_K, \mathbf{x}'_{\tilde{C}}]. \end{aligned}$$

Combining both inequalities we obtain the result. $\square$

The general bound seems to imply that changing a single record has a completely deterministic effect on the remaining correlated records. However, the *strength* of the correlation in a real world scenario may not be that pronounced. Therefore, we investigate the impact of limiting the Pearson correlation coefficient ρ . Theorem 5.3 shows that the general bound is tight, even if we limit the number of correlated records to the minimum, $m = 2$, and $\rho > 0$ is limited to be arbitrarily small.

Theorem 5.3. *Given a random vector $\mathbf{X} = (X_1, X_2)$ and an ε -DP mechanism $\mathcal{M}: \{0, 1\}^2 \rightarrow \mathcal{Y}$ with input data drawn from the distribution of $\mathbf{X}$. Let $S \subseteq \mathcal{Y}$ be a measurable set such that $\mathcal{M}$ reaches the maximum allowed privacy leakage, i.e., there exist two neighboring databases $D, D' \in \mathcal{X}^n$ such that*

$$\Pr[\mathcal{M}(D) \in S] = e^\varepsilon \cdot \Pr[\mathcal{M}(D') \in S].$$

Then, for all $\hat{\rho} > 0$ and for all $\delta > 0$ there exists a probability distribution π with the following property: If the random vector $\mathbf{X}$ follows distribution π , then the Pearson correlation coefficient $\rho_{X_1, X_2} \leq \hat{\rho}$ of the random variables X_1 and X_2 is bounded, and $\text{BDPL} \geq 2\varepsilon - \delta$.

Proof of Theorem 5.3. Let $r \in \mathbb{N}$. Table 2 shows a valid probability distribution π for $\mathbf{X} = (X_1, X_2)$ with $\alpha := \frac{r-1}{r}$ and $\beta := \frac{1}{r^2}$.

	$X_1 = 0$	$X_1 = 1$	
$X_2 = 0$	β	$1 - \alpha - \beta$	$1 - \alpha$
$X_2 = 1$	0	α	α
	β	$1 - \beta$	1

TABLE 2. Joint probability distribution of random variables X_1 and X_2 .

In order to prove the theorem, it suffices to show that when $r \rightarrow \infty$, the correlation coefficient ρ_{X_1, X_2} converges to zero and the BDPL converges to the general upper bound 2ε .

We first look at the Pearson correlation coefficient. The variances of X_1 and X_2 are $\sigma_{X_1}^2 = \beta(1 - \beta)$ and $\sigma_{X_2}^2 = \alpha(1 - \alpha)$ respectively. Then,

$$\rho_{X_1, X_2} = \frac{\alpha - \alpha(1 - \beta)}{\sqrt{\beta(1 - \beta)\alpha(1 - \alpha)}} = \sqrt{\frac{\alpha\beta}{(1 - \beta)(1 - \alpha)}} = \sqrt{\frac{r - 1}{r^2 - 1}} \xrightarrow{r \rightarrow \infty} 0$$

Moreover, by hypothesis there exist two neighboring databases $D, D' \in \{0, 1\}^2$ such that the ε bound is achieved. Without loss of generality we assume they differ in the first coordinate, otherwise by inverting Table 2 we get the same result. Then, computing the BDPL we obtain

$$\begin{aligned} e^{\text{BDPL}} &\geq \frac{\Pr[Y \in S \mid X_1 = 0]}{\Pr[Y \in S \mid X_1 = 1]} = \frac{\sum_{x_2 \in \{0, 1\}} \Pr[Y \in S \mid X_1 = 0, X_2 = x_2] \Pr[X_2 = x_2 \mid X_1 = 0]}{\sum_{x_2 \in \{0, 1\}} \Pr[Y \in S \mid X_1 = 1, X_2 = x_2] \Pr[X_2 = x_2 \mid X_1 = 1]} \\ &= \frac{e^{2\varepsilon} \Pr[\mathcal{M}(1, 1) \in S] \frac{\beta}{\beta} + e^\varepsilon \Pr[\mathcal{M}(1, 1) \in S] \frac{0}{\beta}}{e^\varepsilon \Pr[\mathcal{M}(1, 1) \in S] \frac{1 - \alpha - \beta}{1 - \beta} + \Pr[\mathcal{M}(1, 1) \in S] \frac{\alpha}{1 - \beta}} \\ &= \frac{e^{2\varepsilon}(1 - \beta)}{e^\varepsilon(1 - \alpha - \beta) + \alpha} = e^{2\varepsilon} \frac{r^2 - 1}{e^\varepsilon(r - 1) + r^2 - r}. \end{aligned}$$

Therefore, when r tends to infinity we have $e^{\text{BDPL}} \geq \lim_{r \rightarrow \infty} e^{2\varepsilon} \frac{r^2 - 1}{e^\varepsilon(r - 1) + r^2 - r} = e^{2\varepsilon}$, concluding the result. $\square$

We can choose δ arbitrarily close to zero, for instance we can take the sequence $\delta_n = \{\frac{1}{n}\}_{n \in \mathbb{N}}$, and—independent of the Pearson correlation coefficient—we have

$$\text{BDPL} \geq \lim_{n \rightarrow \infty} (2\varepsilon - \delta_n) = 2\varepsilon.$$

Thus, since the general upper bound of the BDPL of an ε -DP mechanism is 2ε we have $\text{BDPL} = 2\varepsilon$. Therefore, Theorem 5.3 shows that, without additional hypotheses, the general bound from Theorem 5.2 is tight, even if we limit Pearson correlation coefficient ρ to be arbitrarily small.

Theorem 5.3 applies to mechanisms where the estimated ε is tight. This assumption is reasonable, otherwise we could simply estimate a smaller ε and apply the theorem.

5.2. Accuracy. The general bound on the BDPL of a DP mechanism allows us to use $(\frac{\varepsilon}{m})$ -DP mechanisms as ε -BDP mechanisms. However, reducing the privacy leakage of a DP mechanism often has a negative impact on utility. In particular, we investigate the impact on the accuracy of the Laplace mechanism when it is calibrated to protect data with a fixed number of correlated records. As a consequence of our result Theorem 5.2 and Proposition 3.5 we obtain the following result:

Corollary 5.4. *Let $\mathcal{M}_{\varepsilon, f}$ be the Laplace ε -DP mechanism that approximates the query $f: \mathcal{X}^n \rightarrow \mathbb{R}$ with input described by the random vector $\mathbf{X} = (X_1, \dots, X_n)$ with at most $m \leq n$ correlated random variables that follows distribution π . Then, if $\mathcal{M}_{\varepsilon, f}$ is (α, β) -accurate w.r.t. f , there exists an ε -BDP mechanism $\mathcal{B}$ whose input is drawn from π and that is $(m\alpha, \beta)$ -accurate w.r.t. f .*

Proof. From Proposition 3.5, we know that the (α, β) -accuracy of $\mathcal{M}_{\varepsilon, f}$ with respect to f is

$$\alpha = \ln\left(\frac{1}{\beta}\right) \cdot \frac{\Delta f}{\varepsilon}$$

for any $\beta \in (0, 1]$ because $\mathcal{M}_{\varepsilon, f}$ uses the Laplace mechanism.

The idea is to also use the Laplace mechanism for $\mathcal{B}$, but to use an adjusted DP privacy parameter ε' so that $\mathcal{B}$ is ε -BDP. We will see that this results in $(m\alpha, \beta)$ -accuracy. With the general bound from Theorem 5.2, we know that the mechanism $\mathcal{B}$ is $m\varepsilon'$ -BDP. Thus, we have

$$m\varepsilon' = \varepsilon \Leftrightarrow \varepsilon' = \frac{1}{m}\varepsilon.$$

Now, we can calculate the accuracy of mechanism $\mathcal{B}$ because it also uses the Laplace mechanism and we now know the used DP privacy leakage $\varepsilon' = \varepsilon/m$. Mechanism $\mathcal{B}$ is (α', β) -accurate with

$$\alpha' = \ln\left(\frac{1}{\beta}\right) \cdot \frac{\Delta f}{\varepsilon'} = \ln\left(\frac{1}{\beta}\right) \cdot \frac{m\Delta f}{\varepsilon} = m\alpha.$$

Thus, mechanism $\mathcal{B}$ is $(m\alpha, \beta)$ -accurate. $\square$

This result shows that the error α of the Laplace mechanism increases proportionally with the number of correlated records when moving from ε -DP to ε -BDP, and while making no assumption about the distribution of the records. This may be acceptable when the number of correlated records m is small. For example, if $m = 2$, the error α doubles when transitioning from DP to BDP. If the DP mechanism's error is small, this increase may be acceptable. However, utility sharply decreases as m grows.

Since we have shown that our bound on BDPL is tight under the assumption of arbitrary correlation, the utility bound cannot be improved, even when the Pearson correlation coefficient is close to zero. This motivates the next two sections, where we investigate whether additional assumptions on the correlation model can lead to tighter bounds, enabling reduced noise and improved utility while still protecting against correlation attacks.

6. MULTIVARIATE GAUSSIAN CORRELATION

A wide variety of phenomena are effectively modeled using a Gaussian distribution [32]. For example, physiological measures such as height and weight are correlated among family members, and the joint distribution of height and weight in a large population is well fit by a bivariate Gaussian distribution [5]. Consequently, we explore the applicability of BDP to multivariate Gaussian data.

When we are dealing with a database of n records, and each record is drawn from a Gaussian distribution, we can model the joint distribution of all records as a multivariate Gaussian distribution. This model also captures linear correlation between records [35].

Definition 6.1 (Multivariate Gaussian Distribution [35]). Let $\mathbf{X} = (X_1, \dots, X_n)$ be a random vector, let vector $\mu \in \mathbb{R}^n$ be real and let matrix $\Sigma \in \mathbb{R}^{n \times n}$ be symmetric and positive definite. We say $\mathbf{X}$ follows the *multivariate Gaussian distribution with mean μ and covariance Σ* if the probability density of $\mathbf{X}$ for any point $\mathbf{x} \in \mathbb{R}^n$ is

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^\top \Sigma^{-1}(\mathbf{x} - \mu)\right),$$

where $|\Sigma|$ is the determinant of Σ . We write $\mathbf{X} \sim \mathcal{N}(\mu, \Sigma)$.

We establish a relationship between DP and BDP for data drawn from a multivariate Gaussian distribution, based on the maximum Pearson correlation coefficient, which is calculated directly from the covariance matrix [35]. This provides a new, tighter upper bound for the BDPL that improves upon the specific Gaussian bound given in [41] and upon the general bound $n\varepsilon$ for any correlation model.

However, our bound applies only to a specific class of mechanisms: those that satisfy both DP and metric privacy under the ℓ_1 metric. We show in Section 6.2 that the clipped Laplace mechanism meets these criteria and develop a practical application in Section 8.1. To establish this result, we first connect metric privacy with an analogous form of BDP, termed Bayesian metric privacy, which we define below.

6.1. Relationship between Metric Privacy and Bayesian Metric Privacy. Unbounded continuous data domains, such as $\mathbb{R}^n$, usually produce challenges on DP application due to infinity sensitivities [2]. In the context of BDP, Yang et al. [41] defined a relaxation to work in those domains: If the data domain is equivalent to the real numbers (i.e., $\mathcal{X}^n = \mathbb{R}^n$), they defined a modified leakage, $\text{BDPL}(\mathcal{M}; M)$, where they only take into account the leakage between points

with a distance smaller than $M \in \mathbb{R}$, i.e.,

$$\sup_{|x_i - x'_i| \leq M, \mathbf{x}_K, S} \ln \frac{\Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x_i]}{\Pr[Y \in S \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x'_i]}.$$

Applying this BDP relaxation leaves indistinguishability between records at distances greater than M entirely uncontrolled. While this may increase applicability, it reduces privacy and limits insights into the impact of correlation.

However, metric privacy provides a solution to quantify privacy leakage as the distance $d(D, D')$ for each pair of databases D, D' when the maximum privacy leakage cannot be bounded [8]. Therefore, we define Bayesian metric privacy as equivalent to metric privacy where the indistinguishability between two records x, x' depends on the distance $d(x, x')$ between them. Note that the change from databases to records is necessary because BDP does not apply to neighboring databases, but to target records, as we describe in Section 3.2. In this way, we can work with $\mathbb{R}^n$ as the data domain without losing information about the privacy leakage.

Definition 6.2 (Target Dependent Leakage). Given $\mathcal{M} : \mathcal{X}^n \rightarrow \mathcal{Y}$ a randomized mechanism, $\mathbf{X}$ the input random vector following the distribution π , the targeted record index $i \in [n]$, and the known record indices $K \subseteq [n] \setminus \{i\}$, the *adversary-specific target dependent* BDPL of $\mathcal{M}$ w.r.t. adversary (K, i) for any target values $x, x' \in \mathcal{X}$ is

$$\text{BDPL}_{(K,i)}(x, x') = \sup_{\mathbf{x}_K, s} \ln \frac{p_Y(s \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x)}{p_Y(s \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x')}.$$

Given that we understand the leakage for each pair of data records we can simply define Bayesian metric privacy analogously to the original metric privacy notion:

Definition 6.3 (Bayesian Metric Privacy). Let d be a (pseudo)metric on $\mathcal{X}^2$. A mechanism $\mathcal{M}$ is *Bayesian d -private* if for all $x, x' \in \mathcal{X}$,

$$\text{BDPL}(x, x') = \sup_{i, K} \text{BDPL}_{(K,i)}(x, x') \leq \varepsilon,$$

where the supremum is taken over all the possible set of indices $i \in [n]$ and $K \subseteq [n] \setminus \{i\}$. $\text{BDPL}(x, x')$ is called *target dependent* BDPL.

The only difference between BDP and Bayesian d -privacy is that Bayesian d -privacy does not take the supremum over x, x' . Furthermore, Bayesian d -privacy is equivalent to BDP when the distance metric is defined as $d(x, x') = \varepsilon$ for $x \neq x'$ and $d(x, x') = 0$ otherwise.

Now we can prove the relation between a d -private and a Bayesian d -private mechanism when the data distribution is a multivariate Gaussian. Particularly, we focus on the ℓ_1 distance due to its direct application to the Gaussian case. We formalize the conditions needed to obtain our bound:

Definition 6.4. For $\rho \in [0, 1]$ and $n \in \mathbb{N}$, we call the matrix $\Sigma_\rho \in \mathbb{R}^{n \times n}$ a *limited covariance matrix* if

- the matrix Σ_ρ is symmetric and positive definite,
- the diagonal of Σ_ρ is constant, i.e., there is a variance $\sigma^2 > 0$ so that $\Sigma_{\rho, ii} = \sigma^2$ for all $i \in [n]$ and,
- any pairwise correlation is limited by ρ . I.e., for all $i \neq j$ we have $|\Sigma_{\rho, ij}| \leq \rho \sigma^2$.

The first condition is required to be a valid covariance matrix for a Gaussian distribution (see Definition 5.1). The second condition ensures that no records have a deviating variance, i.e., all records are drawn from the same one-dimensional distribution. The final condition imposes that the maximum Pearson correlation coefficient between any two random variables X_i and X_j is bounded by ρ .

If we limit ρ to be small enough, specifically, smaller than $\frac{1}{n-2}$ we get a bound on the BDPL as presented in Theorem 6.6. However, before we can prove Theorem 6.6, we require a lemma that establishes the maximum BDPL of a single adversary.

Lemma 6.5. Let $\mathcal{M}$ with data domain $\mathbb{R}^n$ be an $(\varepsilon\ell_1)$ -private mechanism where $\varepsilon > 0$ and $n \geq 2$ with input data drawn from a multivariate Gaussian distribution $\mathcal{N}(\mu, \Sigma)$.

Let $K = \{n - k, \dots, n - 1\}$ be the set of known indices for the adversary, with $k \leq n - 2$, and $T = K \cup \{n\}$ the known indices including the target. If the principal submatrix Σ_T spanning the last $k + 1$ rows and columns is invertible then the adversary-specific target dependent BDPL of $\mathcal{M}$ for any target values $x_n, x'_n \in \mathbb{R}$ is bounded by

$$\text{BDPL}_{(K,n)}(x_n, x'_n) \leq \varepsilon |x_n - x'_n| \left(\|\Sigma_{U,T} \Sigma_T^{-1} \mathbf{e}_{k+1}\|_1 + 1 \right)$$

where $\mathbf{e}_{k+1} \equiv (0, \dots, 0, 1)^\top \in \mathbb{R}^{k+1}$ and the notation of the Gaussian distribution $\mathcal{N}(\mu, \Sigma)$ is decomposed in

$$\mu = \begin{pmatrix} \mu_U \\ \mu_T \end{pmatrix}, \Sigma = \begin{pmatrix} \Sigma_U & \Sigma_{U,T} \\ \Sigma_{U,T}^\top & \Sigma_T \end{pmatrix}.$$

Proof. The proof is derived from the multivariate Gaussian distribution properties. First, we prove that for any $\mathbf{x}_U \in \mathbb{R}^u$ and $\mathbf{x}_T, \mathbf{x}'_T \in \mathbb{R}^{n-u}$, there exists a $\gamma \in \mathbb{R}^u$, such that

$$p_{\mathbf{x}_U}(\mathbf{x}_U \mid \mathbf{X}_T = \mathbf{x}_T) = p_{\mathbf{x}_U}(\mathbf{x}_U + \gamma \mid \mathbf{X}_T = \mathbf{x}'_T). \quad (6.1)$$

Then, the combination of this property with the $\varepsilon\ell_1$ condition gives the result.

We prove Equation (6.1) by using that the conditional distribution of a Gaussian distribution also follows the Gaussian distribution [33], i.e., $\mathbf{X}_U \mid \mathbf{X}_T = \mathbf{x}_T \sim \mathcal{N}(\hat{\mu}_U, \hat{\Sigma}_U)$ with

$$\hat{\mu}_U = \mu_U + \Sigma_{U,T} \Sigma_T^{-1} (\mathbf{x}_T - \mu_T), \quad \hat{\Sigma}_U = \Sigma_U - \Sigma_{U,T} \Sigma_T^{-1} \Sigma_{U,T}^\top.$$

While the conditional mean $\hat{\mu}_U$ depends on the specific value $\mathbf{x}_T$, the conditional covariance $\hat{\Sigma}_U$ remains fixed. Therefore, the two distributions (conditioned on $\mathbf{x}_T$ and $\mathbf{x}'_T \in \mathbb{R}^{k+1}$ respectively) only differ by a translation, i.e., for any $\mathbf{x}_U \in \mathbb{R}^u$ we have

$$\begin{aligned} p_{\mathbf{x}_U}(\mathbf{x}_U + \Sigma_{U,T} \Sigma_T^{-1} (\mathbf{x}_T - \mathbf{x}'_T) \mid \mathbf{X}_T = \mathbf{x}_T) \\ &= \frac{1}{\sqrt{(2\pi)^u |\hat{\Sigma}_U|}} \exp\left(-\frac{1}{2} (\mathbf{x}_U + \Sigma_{U,T} \Sigma_T^{-1} (\mathbf{x}_T - \mathbf{x}'_T) - \hat{\mu}_U)^\top \hat{\Sigma}_U^{-1} (\mathbf{x}_U + \Sigma_{U,T} \Sigma_T^{-1} (\mathbf{x}_T - \mathbf{x}'_T) - \hat{\mu}_U)\right) \\ &= \frac{1}{\sqrt{(2\pi)^u |\hat{\Sigma}'_U|}} \exp\left(-\frac{1}{2} (\mathbf{x}_U - \hat{\mu}'_U)^\top \hat{\Sigma}'_U^{-1} (\mathbf{x}_U - \hat{\mu}'_U)\right) \\ &= p_{\mathbf{x}_U}(\mathbf{x}_U \mid \mathbf{X}_T = \mathbf{x}'_T). \end{aligned}$$

Therefore, we have shown that the condition of the probability density $p_{\mathbf{x}_U}$ can simply be changed by additively shifting the input for the density, where the shift is $\gamma = \Sigma_{U,T} \Sigma_T^{-1} (\mathbf{x}'_T - \mu_T)$.

Second, we can use the fact that mechanism $\mathcal{M}$ is $(\varepsilon\ell_1)$ -private, therefore, for all $\mathbf{x}_n, \mathbf{x}'_n \in \mathbb{R}^n$,

$$p_Y(s \mid \mathbf{X}_n = \mathbf{x}_n) \leq \exp(\varepsilon \|\mathbf{x}_n - \mathbf{x}'_n\|_1) p_Y(s \mid \mathbf{X}_n = \mathbf{x}'_n) \quad (6.2)$$

Now, applying Equation (6.1) and Equation (6.2) to the density function computation we can bound the adversary-specific target dependent BDPL. Let Y be the random variable that represents the output of mechanism $\mathcal{M}$. Let random vector $\mathbf{X} = (X_1, \dots, X_n)$ follow the multivariate Gaussian distribution $\mathbf{X} \sim \mathcal{N}(\mu, \Sigma)$. To simplify notation, we combine the random vector $\mathbf{X}_K$ and random variable X_n into one target-inclusive vector $\mathbf{X}_T$ with $\mathbf{x}_T = (\mathbf{x}_K, x_n)^\top$ and $\mathbf{x}'_T = (\mathbf{x}_K, x'_n)^\top$. Considering the definition of the adversary-specific target dependent BDPL of (K, n) we have

$$\text{BDPL}_{(K,n)}(x_n, x'_n) = \sup_{\mathbf{x}_K, s} \ln \frac{p_Y(s \mid \mathbf{X}_K = \mathbf{x}_K, X_n = x_n)}{p_Y(s \mid \mathbf{X}_K = \mathbf{x}_K, X_n = x'_n)} \equiv \sup_{\mathbf{x}_T, \mathbf{x}'_T, s} \ln \frac{p_Y(s \mid \mathbf{X}_T = \mathbf{x}_T)}{p_Y(s \mid \mathbf{X}_T = \mathbf{x}'_T)}$$

with the supremum taken over $s \in \mathbb{R}$ and known values $\mathbf{x}_K \in \mathbb{R}^k$.

We calculate the adversary-specific target dependent BDPL by rewriting the density $p_Y(s \mid \mathbf{x}_K, x_n)$ in terms of $p_Y(s \mid \mathbf{x}_K, x'_n)$:

$$p_Y(s \mid \mathbf{X}_K = x_K, X_n = x_n) \equiv p_Y(s \mid \mathbf{X}_T = \mathbf{x}_T)$$

$$\begin{aligned}
&= \int_{\mathbb{R}^u} p_Y(s \mid \mathbf{X}_U = \mathbf{x}_U, \mathbf{X}_T = \mathbf{x}_T) p_{\mathbf{X}_U}(\mathbf{x}_U \mid \mathbf{X}_T = \mathbf{x}_T) d\mathbf{x}_U \\
&= \int_{\mathbb{R}^u} p_Y(s \mid \mathbf{X}_U = \tilde{\mathbf{x}}_U + \gamma, \mathbf{X}_T = \mathbf{x}_T) p_{\mathbf{X}_U}(\tilde{\mathbf{x}}_U + \gamma \mid \mathbf{X}_T = \mathbf{x}_T) d\tilde{\mathbf{x}}_U \tag{6.3}
\end{aligned}$$

$$= \int_{\mathbb{R}^u} p_Y(s \mid \mathbf{X}_U = \tilde{\mathbf{x}}_U + \gamma, \mathbf{X}_T = \mathbf{x}_T) p_{\mathbf{X}_U}(\tilde{\mathbf{x}}_U \mid \mathbf{X}_T = \mathbf{x}'_T) d\tilde{\mathbf{x}}_U \tag{6.4}$$

$$\leq \exp((\|\gamma\|_1 + |x_n - x'_n|)\varepsilon) \int_{\mathbb{R}^u} p_Y(s \mid \mathbf{X}_U = \tilde{\mathbf{x}}_U, \mathbf{X}_T = \mathbf{x}'_T) p_{\mathbf{X}_U}(\tilde{\mathbf{x}}_U \mid \mathbf{X}_T = \mathbf{x}'_T) d\tilde{\mathbf{x}}_U \tag{6.5}$$

$$= \exp((\|\gamma\|_1 + |x_n - x'_n|)\varepsilon) p_Y(s \mid \mathbf{X}_K = x_K, X_n = x'_n)$$

We substitute $\mathbf{x}_U$ for $\tilde{\mathbf{x}}_U + \gamma$ in eq. (6.3) using the change of variable theorem for multiple integrals [36, p. 310]. The substitution is linear so the domain $\mathbb{R}^u$ over which we integrate does not change, and the determinant of the Jacobian of the substitution is simply 1. Then, we apply eq. (6.1) to obtain eq. (6.4). Finally, we use the inequality from eq. (6.2) to derive eq. (6.5), since $\|(\mathbf{x}_U, \mathbf{x}_T) - (\mathbf{x}_U + \gamma, \mathbf{x}'_T)\|_1 = \|\gamma\|_1 + |x_n - x'_n|$.

Now, we can formulate the upper bound of the adversary-specific target dependent BDPL for (K, n) for all $x_n, x'_n \in \mathbb{R}$:

$$\begin{aligned}
\text{BDPL}_{(T)}(x_n, x'_n) &\leq (\|\gamma\|_1 + |x_n - x'_n|)\varepsilon \\
&= (\|\Sigma_{U;T}\Sigma_T^{-1}(\mathbf{x}_T - \mathbf{x}'_T)\|_1 + |x_n - x'_n|)\varepsilon \\
&= (\|\Sigma_{U;T}\Sigma_T^{-1}((\mathbf{x}_K, x_n)^\top - (\mathbf{x}_K, x'_n)^\top)\|_1 + |x_n - x'_n|)\varepsilon \\
&= (\|\Sigma_{U;T}\Sigma_T^{-1}(\mathbf{0}, x_n - x'_n)^\top\|_1 + |x_n - x'_n|)\varepsilon \\
&= (\|\Sigma_{U;T}\Sigma_T^{-1}\mathbf{e}_{k+1}\|_1 + 1)|x_n - x'_n|\varepsilon. \quad \square
\end{aligned}$$

Applying previous lemma, when we limit ρ to be small enough, specifically, smaller than $\frac{1}{n-2}$ we get the following bound:

Theorem 6.6. *Let $\mathcal{M}$ with data domain $\mathbb{R}^n$ be an $(\varepsilon\ell_1)$ -private mechanism where $\varepsilon > 0$ and $n \geq 3$ with input data drawn from a multivariate Gaussian distribution $\mathcal{N}(\mu, \Sigma_\rho)$ with mean $\mu \in \mathbb{R}^n$ and limited covariance matrix $\Sigma_\rho \in \mathbb{R}^{n \times n}$ (Def. 6.4) such that $\rho < \frac{1}{n-2}$ is the maximum correlation coefficient. Then, for any $x, x' \in \mathbb{R}$ we have*

$$\text{BDPL}(x, x') \leq \left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1 \right) |x' - x|.$$

Proof. To prove an upper bound for the target dependent BDPL, we must bound the adversary-specific target dependent BDPL of every possible adversary (K, i) with $i \in [n]$ and $K \subseteq [n] \setminus \{i\}$. The remaining indices besides K and i make up the unknown indices $U = [n] \setminus \{K, i\}$. Lemma 6.5 requires that there is at least one unknown index in U . Thus, we differentiate between two cases.

Case 1: There are no unknown indices besides the target $i \in [n]$, i.e., we have $U = \emptyset$.

Here, the set of known indices $K = [n] \setminus \{i\}$ must contain every index except the target. Therefore, we can calculate the target dependent BDPL of this adversary by using the fact that $\mathcal{M}$ is $(\varepsilon\ell_1)$ -private.

$$\begin{aligned}
\text{BDPL}_{(K,i)}(x_i, x'_i) &= \sup_{\mathbf{x}_K, s} \ln \frac{p_Y(s \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x_i)}{p_Y(s \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x'_i)} \\
&= \sup_{\mathbf{x}_K, s} \ln \frac{p_Y(s \mid \mathbf{X} = D)}{p_Y(s \mid \mathbf{X} = D')} \\
&\leq \sup_{\mathbf{x}_K, s} \ln \frac{e^{\varepsilon\|\mathbf{D}-\mathbf{D}'\|_1} p_Y(s \mid \mathbf{X} = D')}{p_Y(s \mid \mathbf{X} = D')} \\
&= \ln e^{\varepsilon|x_i - x'_i|} = \varepsilon|x_i - x'_i|
\end{aligned}$$

$$\leq \left(\frac{n^2}{4(\frac{1}{\rho} - n + 3)} + 1 \right) \varepsilon |x_i - x'_i|.$$

Case 2: There is at least one unknown record, i.e., $U \neq \emptyset$.

Let $k = |K|$ be the number of known records and $u = |U|$ the number of unknown records. With out loss of generality we have $K = \{n - k, \dots, n - 1\}$ and $i = n$. Otherwise, we simply reorder the components of random vector $\mathbf{X}$ so that the statements about K and i apply.

Choose $\Sigma_U \in \mathbb{R}^{u \times u}$, $\Sigma_{U;T} \in \mathbb{R}^{u \times (k+1)}$ and $\Sigma_T \in \mathbb{R}^{(k+1) \times (k+1)}$ so that the following holds:

$$\Sigma_\rho = \begin{pmatrix} \Sigma_U & \Sigma_{U;T} \\ \Sigma_{U;T}^\top & \Sigma_T \end{pmatrix}$$

In order to use Lemma 6.5, we require the principal submatrix Σ_T spanning the last $k + 1$ rows and columns to be invertible. We show that this is the case by finding a strictly positive lower bound for the eigenvalues of Σ_T . Conveniently, we can later use this fact to bound the entries of Σ_T^{-1} . We denote the individual cells of Σ_T as a_{jl} for all $j, l \in [k + 1]$.

According to the Gershgorin circle theorem [18], every eigenvalue of a real matrix such as Σ_T is contained in a closed disc on the complex number plane with center a_{jj} and radius $\sum_{l \neq j} |a_{jl}|$ for $j \in [k + 1]$. Since the eigenvalues of a symmetric matrix must be real [19, p. 1], we are only concerned with the real part of this disc, i.e., the interval $[a_{jj} - \sum_{l \neq j} |a_{jl}|, a_{jj} + \sum_{l \neq j} |a_{jl}|]$. We can construct a lower bound of the smallest eigenvalue λ_- of Σ_T by finding the lowest border of these intervals.

$$\lambda_- \geq \min_j a_{jj} - \sum_{l \neq j} |a_{jl}| \quad (6.6)$$

$$\geq \min_j \sigma^2 - \sum_{l \neq j} |\rho \sigma^2| \quad (6.7)$$

$$= \sigma^2 - k \rho \sigma^2 \quad (6.8)$$

$$= (1 - k \rho) \sigma^2 \quad (6.9)$$

$$> (1 - (n - 2) \frac{1}{n - 2}) \sigma^2 = 0 \quad (6.10)$$

In eq. (6.7), we use the fact that every random variable has the same variance σ^2 and the correlation between any two random variables in $\mathbf{X}$ is in the interval $[-\rho, \rho]$. Then we show in eq. (6.10) that the bound is positive since the number of known records k must be $n - 2$ or smaller because there is one targeted record and $U \neq \emptyset$ and the maximum correlation ρ is bounded with $\rho < \frac{1}{n-2}$. Thus, we have shown that each eigenvalue of Σ_T is strictly positive and the matrix is therefore invertible.

Now, by direct application of Lemma 6.5 we have an upper bound of the adversary-specific target dependent BDPL for any $x_i, x'_i \in \mathbb{R}$ with

$$\text{BDPL}_{(K,i)}(x_i, x'_i) \leq (\|\Sigma_{U;T} \Sigma_T^{-1} \mathbf{e}_{k+1}\|_1 + 1) \varepsilon |x'_i - x_i|. \quad (6.11)$$

This bound depends on the adversary-specific matrices $\Sigma_{U;T}$ and Σ_T . Our goal is to find a bound for the total target dependent BDPL, irrespective of the specific adversary. We denote the individual cells of $\Sigma_{U;T}$ as b_{jl} for all $j \in [u]$, $l \in [k + 1]$ and the cells of Σ_T^{-1} as α_{jl} for all $j, l \in [k + 1]$.

Since $\Sigma_{U;T}$ only contains covariances between random variables from U and K or n and the covariance is bounded by $\rho \sigma^2$, for all indices $j, l \in [u]$ we obtain that

$$|b_{jl}| \leq \rho \sigma^2. \quad (6.12)$$

Now, we bound the entries of the inverse matrix Σ_T^{-1} . The 2-norm of any symmetric matrix $A \in \mathbb{R}^{m \times m}$ is defined as

$$\|A\|_2 := \sup_{\mathbf{x} \in \mathbb{R}^m} \{\|A\mathbf{x}\|_2 : \|\mathbf{x}\|_2 = 1\}.$$

The 2-norm of A is equal to its maximum singular value (i.e., the largest absolute eigenvalue for a symmetric matrix) [38, p. 47]. Thus, we can use the 2-norm to bound the entries of a

symmetric matrix by its largest absolute eigenvalue $|\lambda_+|$. Let $\mathbf{e}_j \in \mathbb{R}^m$ be the vector with 1 at position j and 0 elsewhere. For every entry a_{ij} in A , we have

$$|a_{ij}| = \sqrt{a_{ij}^2} \leq \sqrt{\sum_{k \in [m]} a_{kj}^2} = \|\mathbf{A}\mathbf{e}_j\|_2 \leq \|\mathbf{A}\|_2 = |\lambda_+|. \quad (6.13)$$

Additionally, the eigenvalues of an inverse A^{-1} can be determined knowing the eigenvalues of A . Let $\lambda \in \mathbb{R}$ be any eigenvalue of A ; λ cannot be zero because A is invertible, consequently,

$$\begin{aligned} Ax &= \lambda x \\ \Leftrightarrow A^{-1}Ax &= \lambda A^{-1}x \\ \Leftrightarrow x &= \lambda A^{-1}x \\ \Leftrightarrow \frac{1}{\lambda}x &= A^{-1}x \end{aligned}$$

Thus, the eigenvalues of A^{-1} are the inverses of the eigenvalues of A . Putting it all together, the entries of Σ_T^{-1} are smaller or equal to the inverse of the smallest absolute eigenvalue of Σ_T . In eq. (6.9), we have shown that all eigenvalues of Σ_T are positive and larger than $(1 - k\rho)\sigma^2$. Therefore, the smallest *absolute* eigenvalue of Σ_T , denoted λ_- is also larger than this bound. Now, these two facts (the entries of a matrix can be bounded by the largest absolute eigenvalue, and the eigenvalues of A^{-1} are the inverses of the eigenvalues of A) are brought together to bound the entries of Σ_T^{-1} :

$$\alpha_{jl} \leq \frac{1}{\lambda_-} \leq \frac{1}{(1 - k\rho)\sigma^2} \quad (6.14)$$

With the bounds for the entries of $\Sigma_{U;T}$ and Σ_T^{-1} in hand, we can find a general upper bound for the adversary-specific target dependent BDPL for any $x_i, x'_i \in \mathbb{R}$.

$$\text{BDPL}_{(K,i)}(x_i, x'_i) \leq (\|\Sigma_{U;T}\Sigma_T^{-1}\mathbf{e}_{k+1}\|_1 + 1)|x'_i - x_i|\varepsilon \quad (6.15)$$

$$= (\|\Sigma_{U;T}(\alpha_{1,k+1}, \dots, \alpha_{k+1,k+1})^\top\|_1 + 1)|x'_i - x_i|\varepsilon \quad (6.16)$$

$$= \left(\left\| \left(\sum_{l=1}^{k+1} \alpha_{l,k+1} b_{1,l}, \dots, \sum_{l=1}^{k+1} \alpha_{l,k+1} b_{u,l} \right)^\top \right\|_1 + 1 \right) |x'_i - x_i|\varepsilon \quad (6.17)$$

$$= \left(\sum_{j=1}^u \left| \sum_{l=1}^{k+1} \alpha_{l,k+1} b_{j,l} \right| + 1 \right) |x'_i - x_i|\varepsilon \quad (6.18)$$

$$\leq \left(\sum_{j=1}^u \left| \sum_{l=1}^{k+1} \frac{\rho\sigma^2}{(1 - k\rho)\sigma^2} \right| + 1 \right) |x'_i - x_i|\varepsilon \quad (6.19)$$

$$= \left(\frac{u(k+1)\rho}{1 - k\rho} + 1 \right) |x'_i - x_i|\varepsilon \quad (6.20)$$

$$= \left(\frac{u(k+1)}{\frac{1}{\rho} - k} + 1 \right) |x'_i - x_i|\varepsilon \quad (6.21)$$

$$\leq \left(\frac{\left(\frac{n}{2}\right)^2}{\frac{1}{\rho} - n + 2} + 1 \right) |x'_i - x_i|\varepsilon \quad (6.22)$$

$$= \left(\frac{n^2}{4\left(\frac{1}{\rho} - n + 2\right)} + 1 \right) |x'_i - x_i|\varepsilon \quad (6.23)$$

We first use the inequality from Lemma 6.5 in eq. (6.15). Then, we multiply Σ_T^{-1} and $\mathbf{e}_{k+1}$; only the last column of Σ_T^{-1} remains in eq. (6.16). The result is multiplied with $\Sigma_{U;T}$ in eq. (6.17). Afterwards, in eq. (6.18) we apply the ℓ_1 -distance to the remaining vector. The bounds for the entries α and b are used in eq. (6.19). Keep in mind that the denominator is always positive

because $k \leq n - 2$ and $\rho < \frac{1}{n-2}$. Then the two sums can be simplified by multiplying by their cardinality in eq. (6.20) since the entries of the sum no longer depend on j or l . Finally, we bound the numerator and denominator in eq. (6.22): The numerator $u(k+1)$ is smaller or equal to $(n/2)^2$ because u and $k+1$ are positive and together form $n = u + k + 1$. Thus, their product is maximal if they meet at the exact midpoint to n . As mentioned previously, the denominator is always positive. It therefore becomes minimal (and the entire expression maximal) if k becomes maximal. This is the case for $k = n - 2$.

In both cases we were able to bound adversary-specific target dependent BDPL as required. Thus, the proof is complete. $\square$

This theorem provides a concrete expression for the increase in privacy loss due to the effect of linear correlation, compared to the independent data case. The greater the Pearson correlation coefficient, the higher the leakage. Additionally, thanks to the relationship between Bayesian metric privacy and BDP, this result allows us to derive a connection between DP and BDP.

6.2. Relationship between DP and BDP.

Observe that any d -private mechanism is an ε -DP mechanism with $\varepsilon = \sup_{D \sim D'} d(D, D')$. Moreover, any Bayesian d -private mechanism is an ε -BDP mechanism with $\varepsilon = \sup_{x, x'} d(x, x')$. By leveraging these relationships between privacy notions we can establish a connection between DP and BDP. However, since this supremum may be unbounded, it can lead to undesirable privacy guarantees. To manage this relationship effectively we apply clipping techniques, resulting in Theorem 6.9, which enables the construction of BDP mechanisms from DP mechanisms. Formally, clipping is defined as:

Definition 6.7. For any interval $I = [a, b] \subset \mathbb{R}$, we define the *clipping function* $c_I : \mathbb{R}^n \rightarrow \mathbb{R}^n$, which, for all $D \in \mathbb{R}^n$ and all $i \in [n]$, outputs

$$c_I(D)_i = \max(a, \min(b, D_i)).$$

Let $\mathcal{M} : \mathbb{R}^n \rightarrow \mathbb{R}$ be a mechanism. We define its *clipped version* $\mathcal{M}_I : \mathbb{R}^n \rightarrow \mathbb{R}$ as $\mathcal{M}_I = \mathcal{M} \circ c_I$.

Due to the data domain reduction, we can bound the DP leakage of $\varepsilon\ell_1$ -private mechanisms.

Lemma 6.8. *If $\mathcal{M} : \mathbb{R}^n \rightarrow \mathbb{R}$ is $\varepsilon\ell_1$ -private, then its clipped version $\mathcal{M}_I$ is $\varepsilon\ell_1$ -private and $(M\varepsilon)$ -DP with $M = |b - a|$.*

Proof. We begin by showing that $\mathcal{M}_I$ is $\varepsilon\ell_1$ -private. Let $D_1, D_2 \in \mathbb{R}^n$ be arbitrary and $S \subseteq \mathbb{R}$ be any measurable set. We have

$$\Pr[\mathcal{M}_I(D_1) \in S] = \Pr[\mathcal{M}(c_I(D_1)) \in S] \tag{6.24}$$

$$\leq e^{\varepsilon \|c_I(D_1) - c_I(D_2)\|_1} \Pr[\mathcal{M}(c_I(D_2)) \in S] \tag{6.25}$$

$$\leq e^{\varepsilon \|D_1 - D_2\|_1} \Pr[\mathcal{M}(c_I(D_2)) \in S] \tag{6.26}$$

$$= e^{\varepsilon \|D_1 - D_2\|_1} \Pr[\mathcal{M}_I(D_2) \in S]. \tag{6.27}$$

In eq. (6.25) we use that $\mathcal{M}$ is $\varepsilon\ell_1$ -private. Then, we apply the fact that the ℓ_1 -distance of two clipped data sets is smaller or equal to the ℓ_1 -distance of the original data sets in eq. (6.26). Finally, eq. (6.27) shows that $\mathcal{M}_I$ is $\varepsilon\ell_1$ -private.

Now, we will show that $\mathcal{M}_I$ is also DP. Let $D, D' \in \mathbb{R}^n$ be arbitrary neighboring data sets, i.e., there only exists a single index $i \in [n]$ with $D_i \neq D'_i$. For any measurable set $S \subseteq \mathcal{Y}$ we have

$$\Pr[\mathcal{M}_I(D) \in S] = \Pr[\mathcal{M}(c_I(D)) \in S] \tag{6.28}$$

$$\leq e^{\varepsilon \|c_I(D) - c_I(D')\|_1} \Pr[\mathcal{M}(c_I(D')) \in S] \tag{6.29}$$

$$= e^{\varepsilon \sum_{j \in [n]} |\max(a, \min(b, D_j)) - \max(a, \min(b, D'_j))|} \Pr[\mathcal{M}(c_I(D')) \in S] \tag{6.30}$$

$$= e^{\varepsilon |\max(a, \min(b, D_i)) - \max(a, \min(b, D'_i))|} \Pr[\mathcal{M}(c_I(D')) \in S] \tag{6.31}$$

$$\leq e^{\varepsilon(b-a)} \Pr[\mathcal{M}(c_I(D')) \in S] = e^{\varepsilon(b-a)} \Pr[\mathcal{M}_I(D') \in S]. \tag{6.32}$$

Once again, we use that $\mathcal{M}$ is $\varepsilon\ell_1$ -private in eq. (6.29). We expand the definition of the ℓ_1 -distance and of the clipping function c_I in eq. (6.30). Then, we use for eq. (6.31) that D and D' only differ for index i . Finally, we can bound the difference between the two entries because they are clipped to the interval $[a, b]$. We have shown that $\mathcal{M}_I$ is $(b - a)\varepsilon$ -DP. $\square$

With Lemma 6.8 and Theorem 6.6, we can directly show that this class of DP-mechanisms has a limited BDPL.

Theorem 6.9 (The Gaussian Bound). *Let $\mathcal{M}_I$ with data domain $\mathbb{R}^n$ be the clipped version of an $(\varepsilon\ell_1)$ -private mechanism $\mathcal{M}$ where $\varepsilon > 0$ and $n \geq 3$ with input data drawn from a multivariate Gaussian distribution $\mathcal{N}(\mu, \Sigma_\rho)$ with mean $\mu \in \mathbb{R}^n$ and limited covariance matrix $\Sigma_\rho \in \mathbb{R}^{n \times n}$ (Def. 6.4) such that $\rho < \frac{1}{n-2}$ is the maximum correlation coefficient. Then, the clipped mechanism $\mathcal{M}_I$ is*

$$\left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1 \right) M\varepsilon\text{-BDP}.$$

where M is the diameter of the interval I .

Proof. We know that $\mathcal{M}_I$ is a $\varepsilon\ell_1$ -private query because of Lemma 6.8. Thus, we can apply Theorem 6.6 to find a universal bound of the target dependent BDPL in this situation. Therefore, the idea is to show how the BDPL is bounded by the target dependent BDPL, and to then apply Theorem 6.6.

$$\text{BDPL} := \sup_{K,i} \text{BDPL}_{(K,i)} \quad (6.33)$$

$$= \sup_{K,i} \left(\sup_{\mathbf{x}_K, s, |x_i - x'_i| \leq M} \ln \frac{p_Y(s \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x_i)}{p_Y(s \mid \mathbf{X}_K = \mathbf{x}_K, X_i = x'_i)} \right) \quad (6.34)$$

$$= \sup_{K,i} \left(\sup_{|x_i - x'_i| \leq M} \text{BDPL}_{(K,i)}(x_i, x'_i) \right) \quad (6.35)$$

$$= \sup_{|x_i - x'_i| \leq M} \left(\sup_{K,i} \text{BDPL}_{(K,i)}(x_i, x'_i) \right) \quad (6.36)$$

$$= \sup_{|x_i - x'_i| \leq M} \text{BDPL}(x_i, x'_i) \quad (6.37)$$

$$\leq \sup_{|x_i - x'_i| \leq M} \left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1 \right) |x'_i - x_i| \varepsilon \quad (6.38)$$

$$= \sup_{|x_i - x'_i| \leq M} \left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1 \right) |x_i - x'_i| \varepsilon \quad (6.39)$$

$$= \left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1 \right) M\varepsilon. \quad (6.40)$$

In eqs. (6.33) and (6.34) we expand the definition of BDPL. Then, in eq. (6.35) we use that the adversary-specific *target dependent* BDPL is defined equivalently, except it does not take the supremum over x_i, x'_i . We switch the order of the suprema in eq. (6.36) to subsequently plug in the definition of the general target dependent BDPL. Then, we use Theorem 6.6 to derive eq. (6.38). Finally, the last supremum can be resolved by writing out $d(x_i, x'_i)$ and bounding it with $M\varepsilon$. It follows that clipped mechanism $\mathcal{M}$ is BDP since the BDPL is limited. $\square$

This theorem allows us to systematically build a BDP mechanism by recalibrating the noise of a DP mechanism when the Pearson correlation coefficient ρ is smaller than $\frac{1}{n-2}$. For instance, given the Laplace mechanism $\mathcal{M}$ that adds noise to a data point $x \in \mathbb{R}$ following $\text{Lap}(\frac{M}{\tau})$, where

$$\tau = \varepsilon \frac{4(\frac{1}{\rho} - n + 2)}{n^2 + 4(\frac{1}{\rho} - n + 2)}$$

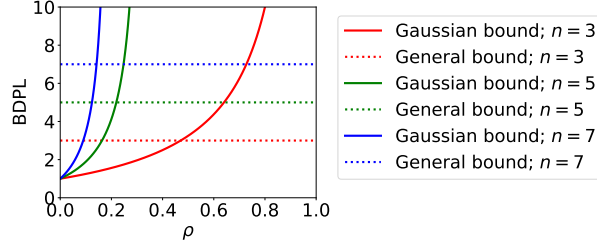


FIGURE 1. General bound compared to Gaussian bound. The x-axis shows the maximum Pearson correlation coefficient ρ . The y-axis is the BDPL. The solid lines show the Gaussian bound from Theorem 6.9 for different database sizes n , the dashed lines are the general bound $n\varepsilon$ of a 1-DP mechanism.

and using the clipped version $\mathcal{M}_I$, we obtain an ε -BDP mechanism.

Notably, for number of records $n \geq 3$ and a maximum correlation coefficient $0 < \rho < \frac{1}{n-2}$ we have

$$\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1 \leq n \text{ if and only if } \rho \leq \frac{n-1}{\frac{5}{4}n^2 - 3n + 2}. \quad (6.41)$$

Consequently, the Gaussian bound improves on the general bound if the Pearson correlation coefficient ρ is restricted to be in the order of $\rho \approx \frac{1}{n}$. This condition is illustrated in Figure 1, where we compare both bounds for clipped mechanisms $\mathcal{M}_I$ that satisfy 1-DP across various database sizes n . We see that the Gaussian bound improves for small ρ values while the general bound is still smaller for larger ones. The higher the number of records n , the better the relative improvement of the Gaussian specific bound compared to the general bound for small correlation ρ .

Importantly, Yang, Sato, and Nakagawa [41] establish a bound for Gaussian Markov random fields. They establish that a clipped $\mathcal{M}_{\varepsilon,f}$ satisfies $(nM\varepsilon)$ -BDP. Theorem 6.9 applies to this particular case since a Gaussian Markov random field is an example of Gaussian Multivariate distribution. Moreover, our bound improves over theirs in the same cases it improves over the general bound.

6.3. Accuracy. When the Pearson correlation is bounded as specified in Equation (6.41), it is guaranteed that a larger ε' than $\frac{\varepsilon}{m}$ is sufficient to guarantee ε -BDP via an ε' -DP mechanism. Since a larger privacy budget generally correlates with improved utility, we can therefore anticipate enhanced utility results. In particular, we express the accuracy improvement of the Laplace mechanism when it is calibrated to protect data drawn from a multivariate Gaussian distribution. As a consequence of our Theorem 6.9 and Proposition 3.5 from [15] we obtain the following result:

Corollary 6.10. *Let $\mathcal{M}_{\varepsilon,f_I}$ be the clipped Laplace ε -DP mechanism that approximates the query f_I as defined in 6.7 with input data of size $n \geq 3$, drawn from a multivariate Gaussian distribution $\mathcal{N}(\mu, \Sigma_\rho)$ with mean $\mu \in \mathbb{R}^n$ and limited covariance $\Sigma_\rho \in \mathbb{R}^{n \times n}$ such that $\rho < \frac{1}{n-2}$. Then, if the Laplace mechanism $\mathcal{M}_{\varepsilon,f_I}$ is (α, β) -accurate w.r.t. f_I , there exists an ε -BDP mechanism $\mathcal{B}$ whose input is drawn from π and that is $(h\alpha, \beta)$ -accurate w.r.t. f_I with*

$$h = \frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1.$$

Proof. The idea of this proof is to construct mechanism $\mathcal{B}$ with the Laplace mechanism as well, but to choose a carefully selected privacy leakage $\varepsilon' < \varepsilon$ so that mechanism $\mathcal{B}$ is (1) ε -BDP and (2) $(h\alpha, \beta)$ -accurate.

First, we determine the accuracy of mechanism $\mathcal{M}_{\varepsilon,f_I}$. According to Proposition 3.5, the (α, β) -accuracy of the Laplace mechanism for a given probability $\beta \in (0, 1]$ and privacy parameter ε is

$$\alpha = \ln\left(\frac{1}{\beta}\right) \frac{\Delta f_I}{\varepsilon}. \quad (6.42)$$

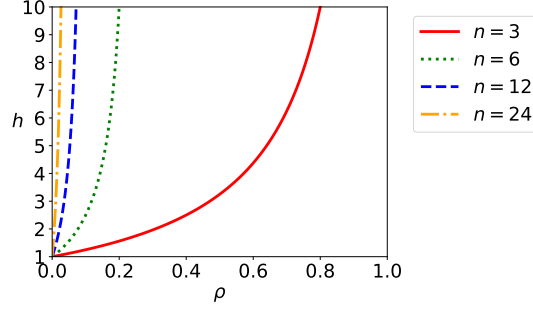


FIGURE 2. Relative accuracy of an ε -BDP mechanism to an ε -DP mechanism for a Multivariate Gaussian distribution. The x-axis shows Pearson correlation ρ , while the y-axis shows factor h . Given Gaussian data with correlation ρ , each line indicates the maximum error increase h on a database of size n for any $\varepsilon > 0$.

So this is the (α, β) -accuracy of $\mathcal{M}_{\varepsilon, f_I}$. We have to show that there exists an ε -BDP mechanism $\mathcal{B}$ which is $(h\alpha, \beta)$ -accurate. We choose $\mathcal{B}$ as the Laplace mechanism applied to f_I with an adjusted privacy parameter $\varepsilon' > 0$. Thus, $\mathcal{B}$ will be ε' -DP. Observe that $\mathcal{B}$ is d -private with $d(D, D') = \frac{\varepsilon'}{M} \|D - D'\|_1$. Therefore, we can use Theorem 6.9 to show that $\mathcal{B}$ is BDP. We must choose ε' in a way that ensures that the BDPL is limited to ε , so that we have ε -BDP. With Theorem 6.9, $\mathcal{B}$ is

$$\left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1\right)\varepsilon'\text{-BDP}. \quad (6.43)$$

Therefore, to achieve ε -BDP, we must have

$$\left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1\right)\varepsilon' = \varepsilon \quad \Leftrightarrow \quad \varepsilon' = \varepsilon \left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1\right)^{-1}. \quad (6.44)$$

Now, we can calculate the accuracy of $\mathcal{B}$ because it also uses the Laplace mechanism. Then, we find an upper bound for this accuracy. Mechanism $\mathcal{B}$ is (α', β) -accurate, with

$$\alpha' = \ln\left(\frac{1}{\beta}\right) \frac{\Delta f_I}{\varepsilon'} = \ln\left(\frac{1}{\beta}\right) \frac{\Delta f_I}{\varepsilon} \left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1\right) \quad (6.45)$$

$$= \alpha \left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1\right) \quad (6.46)$$

$$= \alpha h \quad (6.47)$$

Finally, we find that $\mathcal{B}$ is $(h\alpha, \beta)$ -accurate. $\square$

The statement of Corollary 6.10 is visualized in Figure 2. This figure shows that in order to provide similar utility to DP, ρ must be small. The larger the size n of the database of correlated records, the smaller ρ has to be to provide similar utility.

The results of this section enable the protection of weakly correlated data drawn from a multivariate Gaussian distribution, while improving utility with respect to the general bound and the state of the art [41].

7. MARKOV CHAIN CORRELATION MODEL

In streaming processes or time series data, states at successive time steps are often correlated, meaning that the state at a given time step depends on the state at the previous one. For example, a user's location at time step t is correlated with their location at $t-1$. This dependency pattern is commonly modeled using Markov chains [3].

Consequently, in this section we investigate the impact of the Markov model on the privacy leakage and utility of BDP mechanisms. Particularly, we prove Theorem 7.5, a new bound on

the BDPL of any ε -DP mechanism when data follows a Markov chain. Additionally, we use our results to elaborate on the utility gain compared to protecting against arbitrary correlation.

For the rest of this work we use the definition of a Markov chain by [3]. Strictly speaking, it refers to finite, time-homogeneous Markov chains, i.e., Markov chains with finite state spaces whose transition probabilities remain constant for every time step i . They are formalized as follows:

Definition 7.1 (Markov Chain [3]). Let $\mathcal{S}$ be a finite set of possible states of size $s \in \mathbb{N}$ and let $\mathbf{X} = (X_1, \dots, X_n)$ be a random vector. We say $\mathbf{X}$ is a *Markov chain* with transition matrix $P \in \mathbb{R}^{s \times s}$ and initial distribution $w \in \mathbb{R}^s$ if all of the following holds.

- (1) For all states $x, y \in \mathcal{S}$ and all indices $i \in [n - 1]$ we have $\Pr[X_{i+1} = x | X_i = y] = P_{y,x}$.
- (2) For all states $x \in \mathcal{S}$ we have $\Pr[X_1 = x] = w_x$.
- (3) The Markov property: For all indices $i \in [n - 1]$ and for all states $x_1, \dots, x_i, x_{i+1} \in \mathcal{S}$ we have

$$\begin{aligned} \Pr[X_{i+1} = x_{i+1} \mid X_1 = x_1, \dots, X_i = x_i] \\ = \Pr[X_{i+1} = x_{i+1} \mid X_i = x_i]. \end{aligned}$$

Note that the Markov property from Definition 7.1 holds not only when the full history is known, but also when only the partial history is known as we show in the following remark.

Remark 7.2. Given an index $i \in [n - 1]$, and a set $A \subseteq \{1, \dots, i - 1\}$ containing only indices smaller than i . Then, for any states $x, y \in \mathcal{S}$ and any state tuple $\mathbf{x}_A \in \mathcal{S}^{|A|}$, we have

$$\begin{aligned} \Pr[x_{i+1} \mid x_i, \mathbf{x}_A] &= \sum_{\mathbf{x}_B \in \mathcal{S}^b} \Pr[x_{i+1} | x_i, \mathbf{x}_A, \mathbf{x}_B] \Pr[\mathbf{x}_B | x_i, \mathbf{x}_A] \\ &\stackrel{(*)}{=} \sum_{\mathbf{x}_B \in \mathcal{S}^b} \Pr[x_{i+1} \mid x_i] \Pr[\mathbf{x}_B \mid x_i, \mathbf{x}_A] = \Pr[x_{i+1} \mid x_i] \sum_{\mathbf{x}_B \in \mathcal{S}^b} \Pr[\mathbf{x}_B \mid x_i, \mathbf{x}_A] = \Pr[x_{i+1} \mid x_i], \end{aligned}$$

where $B = [n] \setminus (A \cup \{i\})$ is the set of remaining indices, and $b = |B|$. We use the law of total probability to introduce all remaining indices in B . Then, in $(*)$, we apply the Markov property. Finally, $\Pr[x_{i+1} \mid x_i]$ can be factored out of the sum because it no longer depends on $\mathbf{x}_B$. The remaining sum adds up to 1, so we are left with only the condition of the direct predecessor.

7.1. Relationship between DP and BDP. In this subsection, we show that it is possible to obtain a bound on the BDPL of any DP mechanism based on the maximum ratio between the largest and smallest transition probabilities in the Markov chain. The intuition is that if all transition probabilities are similar, changing the random variable X_i from state x_i to state x'_i will have minimal impact on the subsequent time steps of the Markov chain. However, if the transition probabilities differ significantly, this change could have a large effect over many time steps. To prove our main result Theorem 7.5 we need the auxiliary Lemmas 7.3 and 7.4.

Lemma 7.3 (Generalized Markov Property). *Given $\mathbf{X}$ a Markov chain, for all sets of indices $A \in [n] \setminus \{i\}$:*

$$\Pr[x_i \mid \mathbf{x}_A] = \Pr[x_i \mid x_\ell, x_r]$$

where ℓ, r are the nearest indices to i in A both left and right, i.e., $\ell, r \in A$ with $\ell < i$ and $i < r$ so that for all indices $j \in A$ we have $j < \ell$ or $r < j$. Notably, if all indices in A are smaller than i we only need to consider ℓ and if all are above we only need to consider r .

Proof. We derive this statement directly from probability rules and the Markov property. Let $i, j \in [n]$ be indices and $A' \subseteq [n] \setminus \{i, j\}$ be a set of indices so that there exists an index $\ell \in A'$, $|A'| = a$ that is “in between” i and j , i.e., we have $i > \ell > j$ or $i < \ell < j$. If for all states $x_i, x_j \in \mathcal{S}$ and for all state tuples $\mathbf{x}_{A'} \in \mathcal{S}^a$ we have

$$\Pr[x_i \mid \mathbf{x}_{A'}, x_j] = \Pr[x_i \mid \mathbf{x}_{A'}], \quad (7.1)$$

then it follows

$$\Pr[x_i \mid \mathbf{x}_A] = \Pr[x_i \mid x_{i_1}, \dots, x_\ell, x_r, \dots, x_{i_a}] \quad (7.2)$$

$$= \Pr[x_i \mid x_{i_1}, \dots, x_\ell, x_r] \quad (7.3)$$

$$= \Pr[x_i \mid x_\ell, x_r]. \quad (7.4)$$

Where eq. (7.3) holds because for all $i_j > r$, there exists $r \in A'$ such that $i < r < i_j$; therefore we can apply Equation (7.1). Analogously eq. (7.4) holds because for all $i_j < \ell$, there exists $\ell \in \{\ell, r\}$ such that $i_j < \ell < i$.

Consequently, for the rest of the proof we focus on proving Equation (7.1).

We proceed separately for the case in which the “irrelevant” index j is smaller or left, i.e., there exists $\ell \in A$ such that $j < \ell < i$ and for the case in which j is above or right, i.e., exist $r \in A$ such that $i < r < j$.

Case 1: First, we show that Equation (7.1) holds if $A_1 \subseteq \{1, \dots, i-1\}$ and $\ell \in A_1$ such that $j < \ell$ where $\ell \in A_1$ and $j < \ell < i$ so that no other index in A_1 lies between i and ℓ . This means that the set of indices between ℓ and i —defined as $B = \{\ell+1, \dots, i-1\}$ —is disjoint with A_1 .

If B is empty, then eq. (7.1) follows immediately from Remark 7.2 because index ℓ is the direct predecessor of index i , i.e., $\ell = i-1$.

If $B \neq \emptyset$, using the law of total probability, we have

$$\Pr[x_i \mid \mathbf{x}_{A_1}, x_j] = \sum_{\mathbf{x}_B \in \mathcal{S}^b} \Pr[x_i \mid \mathbf{x}_{A_1}, x_j, \mathbf{x}_B] \Pr[\mathbf{x}_B \mid \mathbf{x}_{A_1}, x_j] \quad (7.5)$$

$$= \sum_{\mathbf{x}_B \in \mathcal{S}^b} \Pr[x_i \mid \mathbf{x}_{A_1}, \mathbf{x}_B] \Pr[\mathbf{x}_B \mid \mathbf{x}_{A_1}] = \Pr[x_i \mid \mathbf{x}_{A_1}], \quad (7.6)$$

where Equation (7.6) follows by applying Remark 7.2, since

$$\Pr[\mathbf{x}_B \mid \mathbf{x}_{A_1}, x_j] = \Pr[x_{\ell+1}, \dots, x_{i-1} \mid \mathbf{x}_{A_1}, x_j] \quad (7.7)$$

$$= \Pr[x_{\ell+1}, \dots, x_{i-2} \mid \mathbf{x}_{A_1}, x_j] \Pr[x_{i-1} \mid \mathbf{x}_{A_1}, x_j, x_{\ell+1}, \dots, x_{i-2}] \quad (7.8)$$

$$= \dots$$

$$= \Pr[x_{\ell+1} \mid \mathbf{x}_{A_1}, x_j] \prod_{k=\ell+2}^{i-1} \Pr[x_k \mid \mathbf{x}_{A_1}, x_j, x_{\ell+1}, \dots, x_{k-1}] \quad (7.9)$$

$$= \Pr[x_{\ell+1} \mid \mathbf{x}_{A_1}] \prod_{k=\ell+2}^{i-1} \Pr[x_k \mid \mathbf{x}_{A_1}, x_{\ell+1}, \dots, x_{k-1}] \quad (7.10)$$

$$= \Pr[\mathbf{x}_B \mid \mathbf{x}_{A_1}]. \quad (7.11)$$

In eq. (7.8), we rewrite the joint probability of $\mathbf{x}_B$ as the two parts X_{i-1} and $\mathbf{x}_{B \setminus \{i-1\}}$. This uses the fact that a joint probability can be rewritten using $\Pr[A \cap B \mid C] = \Pr[A \mid C] \Pr[B \mid C, A]$. Here, event A corresponds to conditions $X_{\ell+1} = x_{\ell+1}, \dots, X_{i-2} = x_{i-2}$, event B to condition $X_{i-1} = x_{i-1}$ and event C to conditions $\mathbf{x}_{A_1} = \mathbf{x}_{A_1}, X_j = x_j$. This step is repeatedly used to fully split $\mathbf{x}_B$ into its components and derive eq. (7.9). Then, we use the Remark 7.2 for eq. (7.10): Random variable $X_{\ell+1}$ is conditionally independent of X_j given the direct predecessor X_ℓ with index $\ell \in A$. Similarly, random variable X_k is conditionally independent of X_j , given the direct predecessor X_{k-1} .

Note that, this result can be extended by induction to say that given any set of indices $C \subseteq [\ell+1, \dots, n]$, if $A_1 \subseteq \{1, \dots, i-1\}$, ℓ the biggest index A_1 and $j < \ell$, then

$$\Pr[x_c \mid \mathbf{x}_{A_1}, x_j] = \Pr[x_c \mid \mathbf{x}_{A_1}] \quad (7.12)$$

For $|C| = 1$, we have just proven Equation (7.12). Now we assume it true for all $|C| \leq n-1$ and we prove it for $|C| = n$:

$$\begin{aligned} \Pr[x_c \mid \mathbf{x}_{A_1}, x_j] &= \Pr[x_{c_1}, \dots, x_{c_n} \mid \mathbf{x}_{A_1}, x_j] \\ &= \Pr[x_{c_n} \mid \mathbf{x}_{A_1}, x_j, \mathbf{x}_{C \setminus \{c_n\}}] \Pr[\mathbf{x}_{C \setminus \{c_n\}} \mid \mathbf{x}_{A_1}, x_j] \\ &= \Pr[x_{c_n} \mid \mathbf{x}_{A_1}, \mathbf{x}_{C \setminus \{c_n\}}] \Pr[\mathbf{x}_{C \setminus \{c_n\}} \mid \mathbf{x}_{A_1}] \\ &= \Pr[x_c \mid \mathbf{x}_{A_1}] \end{aligned}$$

where the last equality follows directly from the induction hypothesis since $|C \setminus \{c_n\}| = n - 1$.

Now, we derive that Equation (7.1) also holds for an arbitrary set of indices A , not necessarily all smaller than i , i.e., $A \subseteq [n] \setminus \{i\}$. We can partition A into the indices before i and after i , i.e., $A = A_1 \cup A_2$ where $A_1 = \{i_j \in A : i_j < i\}$ and $A_2 = \{i_j \in A : i_j > i\}$. Then, we have

$$\Pr[x_i | \mathbf{x}_A, x_j] = \Pr[x_i | \mathbf{x}_{A_1}, \mathbf{x}_{A_2}, x_j] := \frac{\Pr[x_i, \mathbf{x}_{A_1}, \mathbf{x}_{A_2}, x_j]}{\Pr[\mathbf{x}_{A_1}, \mathbf{x}_{A_2}, x_j]} \quad (7.13)$$

$$= \frac{\Pr[x_i, \mathbf{x}_{A_1}, x_j] \Pr[\mathbf{x}_{A_2} | x_i, \mathbf{x}_{A_1}, x_j]}{\Pr[\mathbf{x}_{A_1}, x_j] \Pr[\mathbf{x}_{A_2} | \mathbf{x}_{A_1}, x_j]} \quad (7.14)$$

$$= \Pr[x_i | \mathbf{x}_{A_1}, x_j] \frac{\Pr[\mathbf{x}_{A_2} | x_i, \mathbf{x}_{A_1}, x_j]}{\Pr[\mathbf{x}_{A_2} | \mathbf{x}_{A_1}, x_j]} \quad (7.15)$$

$$= \Pr[x_i | \mathbf{x}_{A_1}] \frac{\Pr[\mathbf{x}_{A_2} | x_i, \mathbf{x}_{A_1}]}{\Pr[\mathbf{x}_{A_2} | \mathbf{x}_{A_1}]} \quad (7.16)$$

$$= \Pr[x_i | \mathbf{x}_{A_1}, \mathbf{x}_{A_2}] = \Pr[x_i | \mathbf{x}_A, x_j] \quad (7.17)$$

where Equation (7.16) follows from Equation (7.12).

Case 2: There exists an index $r \in A$ with $i < l < r$. Here, Equation (7.1) is obtained by applying Bayes' rule to reduce the problem to the already proven first case:

$$\Pr[x_i | \mathbf{x}_A, x_j] = \frac{\Pr[x_j | \mathbf{x}_A, x_i] \Pr[x_i | \mathbf{x}_A]}{\Pr[x_j | \mathbf{x}_A]} \quad (7.18)$$

$$= \frac{\Pr[x_j | \mathbf{x}_A] \Pr[x_i | \mathbf{x}_A]}{\Pr[x_j | \mathbf{x}_A]} \quad (7.19)$$

$$= \Pr[x_i | \mathbf{x}_A] \quad (7.20)$$

We use Bayes' theorem in eq. (7.18). The first probability of the numerator is now in the situation of Case 1, since it is the probability of random variable X_j , conditioned on $\mathbf{x}_A$ and X_i with $i < r < j$ and $r \in A$. Equation (7.1) has already been proven for that case, so we apply it to derive eq. (7.19). Finally, eq. (7.20) follows directly by simplifying. $\square$

Lemma 7.4. Let random vector $\mathbf{X} = (X_1, \dots, X_n)$ be a Markov chain with transition probabilities $P \in \mathbb{R}^{s \times s}$ and initial distribution $w \in \mathbb{R}^s$ with the following properties:

- (H1) Every cell of P is positive, i.e., for all $k, l \in \mathcal{S}$ we have $P_{k,l} > 0$.
- (H2) Vector w is an eigenvector of P to the eigenvalue 1, i.e., $wP = w$.

Let $i \in [n]$ be the target index and let sets $U, K \subseteq [n] \setminus \{i\}$ be disjoint, with $[n] = U \cup K \cup \{i\}$ and at least one index in U . Then, for any unknown states $\mathbf{x}_U \in \mathcal{S}^u$, known states $\mathbf{x}_K \in \mathcal{S}^k$ and target states $x_i, x'_i \in \mathcal{S}$ we have

$$\frac{\Pr[\mathbf{X}_U = \mathbf{x}_U | \mathbf{X}_K = \mathbf{x}_K, X_i = x_i]}{\Pr[\mathbf{X}_U = \mathbf{x}_U | \mathbf{X}_K = \mathbf{x}_K, X_i = x'_i]} \leq \left(\frac{\max_{kl} P_{kl}}{\min_{kl} P_{kl}} \right)^4 \equiv \gamma^4.$$

Proof. First, combining Lemma 7.3 and Bayes' rule we obtain that, for all $A \in [n] \setminus \{i\}$:

$$\Pr[x_i | \mathbf{x}_A] = \frac{\Pr[x_r | x_i, x_\ell] \Pr[x_i | x_\ell]}{\Pr[x_r | x_\ell]} = \frac{\Pr[x_r | x_i] \Pr[x_i | x_\ell]}{\Pr[x_r | x_\ell]} \quad (7.21)$$

where 7.21 follows by application of Lemma 7.3 since i is closer to r than ℓ , i.e., $\ell < i < r$.

Second, we use (H1) and (H2) to prove that given $\mathbf{X}$ a Markov chain, for all indexes i, j , not necessarily consecutive, and for all $x_i, x'_i, y_j, y'_j \in \mathcal{S}$,

$$\frac{\Pr[X_i = x_i]}{\Pr[X_i = x'_i]} \leq \gamma, \quad \frac{\Pr[X_i = x_i | X_j = y_j]}{\Pr[X_i = x_i | X_j = y'_j]} \leq \gamma \text{ and } \frac{\Pr[X_i = x_i | X_j = y_j]}{\Pr[X_i = x'_i | X_j = y_j]} \leq \gamma \quad (7.22)$$

We begin by proving $\frac{\Pr[x_i]}{\Pr[x'_i]} \leq \gamma$. First, we show that w – as an eigenvector of P – not only contains the prior probabilities of X_1 , but of any random variable X_i for $i \in [n]$. I.e., the equality

$\Pr[X_i = x_i] = w_{x_i}$ holds for any state $x_i \in \mathcal{S}$ because w is the equilibrium distribution. We proceed by induction, therefore we assume it true for $i - 1$ and prove it for i :

$$\Pr[X_i = x_i] = \sum_{y \in \mathcal{S}} \Pr[X_i = x_i \mid X_{i-1} = y] \Pr[X_{i-1} = y] \quad (7.23)$$

$$= \sum_{y \in \mathcal{S}} P_{y, x_i} w_y \quad (7.24)$$

$$= (wP)_{x_i} \quad (7.25)$$

$$= w_{x_i} \quad (7.26)$$

We apply the law of total probability in eq. (7.23). Then, we use the transition matrix P and the induction hypothesis ($\Pr[X_{i-1} = y] = w_y$) to replace the probabilities with entries of P and w in eq. (7.24). Then, we rewrite the sum as a matrix-vector product in eq. (7.25). Finally, we take advantage of the fact that w is an eigenvector of P in eq. (7.26). With the basis $\Pr[X_1 = x] = w_x$ (which is the definition of w , see Definition 7.1) and this derivation, we prove by induction that the entries of w are equal to the prior probabilities of any random variable X_i .

Now we can bound the entries of w , thereby also bounding the probabilities $\Pr[X_i = x]$. We prove the upper bound $w_x \leq \max_{k, l \in \mathcal{S}} P_{k, l}$ by contradiction; the lower bound $w_x \geq \min_{k, l \in \mathcal{S}} P_{k, l}$ follows analogously. Assume that there exists a state $y \in \mathcal{S}$ so that its prior probability in w is greater than any transition probability, i.e., $w_y > \max_{k, l \in \mathcal{S}} P_{k, l}$. This leads to a contradiction as follows.

$$w_y = (wP)_y \quad (7.27)$$

$$= \sum_{k \in \mathcal{S}} P_{k, y} w_k \quad (7.28)$$

$$\Leftrightarrow (1 - P_{y, y})w_y = \sum_{k \neq y} P_{k, y} w_k \quad (7.29)$$

$$\Leftrightarrow 1 - P_{y, y} = \sum_{k \neq y} \frac{P_{k, y}}{w_y} w_k \quad (7.30)$$

$$< \sum_{k \neq y} w_k \quad (7.31)$$

$$= 1 - w_y \quad (7.32)$$

$$\Leftrightarrow 1 + w_y < 1 + P_{y, y} \quad (7.33)$$

$$\Leftrightarrow w_y < P_{y, y} \quad (7.34)$$

We use that w is an eigenvector in eq. (7.27) and subsequently rewrite the matrix-vector multiplication. In eq. (7.31), we apply the assumption that w_y is greater than any transition probability, so $P_{k, y}/w_y$ must be strictly smaller than one. Equation (7.32) follows because the entries of w must sum to one as w is a probability distribution (see Definition 7.1). Finally, we arrive at the statement $w_y < P_{y, y}$ which is contradictory to our assumption that w_y is bigger than every $P_{k, y}$. Thus, this assumption was false and probability w_y must be smaller or equal to $\max_{k, l \in \mathcal{S}} P_{kl}$ for any state $y \in \mathcal{S}$.

Second, we prove that $\frac{\Pr[x_i | y_j]}{\Pr[x_i | y'_j]} = \frac{P_{y_j, x_i}}{P_{y'_j, x_i}} \leq \gamma$. Let indices $i, j \in [n]$ such that $j < i$ and let states $x_i, y_j, y'_j \in \mathcal{S}$. If random variables X_i and X_j are direct neighbors, i.e., $i = j + 1$, then the probabilities are transition probabilities from matrix P and the bound follows trivially.

In the other case (i.e., $j + 1 < i$), we have

$$\begin{aligned} \Pr[X_i = x_i \mid X_j = y_j] &= \sum_{y_{j+1} \in \mathcal{S}} \Pr[x_i \mid y_j, X_{j+1} = y_{j+1}] \Pr[X_{j+1} = y_{j+1} \mid y_j] \end{aligned} \quad (7.35)$$

$$= \sum_{y_{j+1} \in \mathcal{S}} \Pr[x_i \mid y_{j+1}] \Pr[y_{j+1} \mid y_j] \quad (7.36)$$

$$= \sum_{y_{j+1} \in \mathcal{S}} \Pr[x_i | y_{j+1}] \Pr[y_{j+1} | y'_j] \frac{\Pr[y_{j+1} | y_j]}{\Pr[y_{j+1} | y'_j]} \quad (7.37)$$

$$= \sum_{y_{j+1} \in \mathcal{S}} \Pr[x_i | y'_j, y_{j+1}] \Pr[y_{j+1} | y'_j] \frac{P_{y_j, y_{j+1}}}{P_{y'_j, y_{j+1}}} \quad (7.38)$$

$$\leq \sum_{y_{j+1} \in \mathcal{S}} \Pr[x_i | y'_j, y_{j+1}] \Pr[y_{j+1} | y'_j] \gamma \quad (7.39)$$

$$\leq \gamma \Pr[X_i = x | X_j = y']. \quad (7.40)$$

We use Lemma 7.3 to remove $X_j = x_j$ from the condition of the first probability in eq. (7.38) because $j + 1$ is closer to i . If $i < j$ then the results follow from applying Bayes' rule and the previous case.

Finally, we prove the last inequality from Equation (7.22), in a similar fashion to the one before. Given $x_i, x'_i, y_j \in \mathcal{S}$. If random variables X_i and X_j are direct neighbors (i.e., $j = i - 1$), the ratio can once again be bounded straightforwardly as it only contains probabilities from P .

Otherwise for $j < i - 1$, we have

$$\begin{aligned} & \Pr[X_i = x_i | X_j = y_j] \\ &= \sum_{x_{i-1} \in \mathcal{S}} \Pr[x_i | y_j, x_{i-1}] \Pr[x_{i-1} | y_j] \end{aligned} \quad (7.41)$$

$$= \sum_{x_{i-1} \in \mathcal{S}} \Pr[x_i | x_{i-1}] \Pr[x_{i-1} | y_j] \quad (7.42)$$

$$= \sum_{x_{i-1} \in \mathcal{S}} \Pr[x'_i | x_{i-1}] \frac{\Pr[x_i | x_{i-1}]}{\Pr[x'_i | x_{i-1}]} \Pr[x_{i-1} | y_j] \quad (7.43)$$

$$= \sum_{x_{i-1} \in \mathcal{S}} \Pr[x'_i | x_{i-1}] \Pr[x_{i-1} | y_j] \frac{P_{x_{i-1}, x_i}}{P_{x_{i-1}, x'_i}} \quad (7.44)$$

$$\leq \sum_{x_{i-1} \in \mathcal{S}} \Pr[x'_i | x_{i-1}, y_j] \Pr[x_{i-1} | y_j] \gamma \quad (7.45)$$

$$= \gamma \Pr[X_i = x' | X_j = y]. \quad (7.46)$$

Lemma 7.3 is used to drop $X_j = y$ from the condition in eq. (7.42) and add it again in Equation (7.45). If $i < j$ then the results follow from applying the Bayes rule and the previous case.

Combining Equation (7.21) and Equation (7.22) we obtain that

$$\frac{\Pr[x_i | \mathbf{x}_A]}{\Pr[x'_i | \mathbf{x}_A]} = \frac{\Pr[x_r | x_i] \Pr[x_i | x_\ell]}{\Pr[x_r | x'_i] \Pr[x'_i | x_\ell]} \leq \gamma^2 \quad (7.47)$$

Note that if A only contains indices smaller than i , then previous equation gets simplified to

$$\frac{\Pr[x_i | \mathbf{x}_A]}{\Pr[x'_i | \mathbf{x}_A]} = \frac{\Pr[x_i | x_\ell]}{\Pr[x'_i | x_\ell]} \leq \gamma \leq \gamma^2$$

and the analogous holds if A only contains indices bigger than i .

Finally, combining Bayes' rule with Equation (7.47) applied to $A = K$ and $A = [n] \setminus \{i\}$, we obtain the result:

$$\frac{\Pr[\mathbf{x}_U | \mathbf{x}_K, x_i]}{\Pr[\mathbf{x}_U | \mathbf{x}_K, x'_i]} = \frac{\Pr[x_i | \mathbf{x}_K, \mathbf{x}_U] \Pr[x'_i | \mathbf{x}_K]}{\Pr[x'_i | \mathbf{x}_K, \mathbf{x}_U] \Pr[x_i | \mathbf{x}_K]} = \frac{\Pr[x_i | \mathbf{x}_{-i}] \Pr[x'_i | \mathbf{x}_K]}{\Pr[x'_i | \mathbf{x}_{-i}] \Pr[x_i | \mathbf{x}_K]} \leq \gamma^2 \gamma^2 = \gamma^4$$

Note that if $K = \emptyset$ the previous expression gets simplified to

$$\frac{\Pr[\mathbf{x}_U | x_i]}{\Pr[\mathbf{x}_U | x'_i]} = \frac{\Pr[x_i | \mathbf{x}_U] \Pr[x'_i]}{\Pr[x'_i | \mathbf{x}_U] \Pr[x_i]} \leq \gamma^3 \leq \gamma^4$$

□

Theorem 7.5 (The Markov Chain Bound). *Let $s \in \mathbb{N}$ be the number of states. Let $\mathcal{M} : \mathcal{S}^n \rightarrow \mathcal{Y}$ be an ε -DP mechanism. Let the databases follow a Markov chain with transition matrix $P \in \mathbb{R}^{s \times s}$ and initial distribution $w \in \mathbb{R}^s$ with the following properties:*

- (H1) *For all $x, y \in \mathcal{S}$ we have $P_{x,y} > 0$ and,*
- (H2) *$wP = w$.*

Then, $\mathcal{M}$ is an $(\varepsilon + 4 \ln \gamma)$ -BDP mechanism where

$$\gamma := \frac{\max_{x,y \in \mathcal{S}} P_{xy}}{\min_{x,y \in \mathcal{S}} P_{xy}}.$$

Proof. If there are no unknown indices $U = \emptyset$ the adversary knows every index $K = [n] \setminus \{i\}$ except the target index and the adversary-specific $\text{BDPL}_{(K,i)}$ becomes the same as the DP privacy leakage [41]. Thus, since $\varepsilon \leq \varepsilon + 4\gamma$ for all $\gamma \geq 1$, the inequality is trivially satisfied.

We denote by $u = |U|$ the size of unknown indexes. If there is at least one unknown index for the adversary, i.e., $u \geq 1$ then we have

$$\begin{aligned} \Pr_{\mathcal{M}}[Y \in S | \mathbf{x}_K, x_i] &= \sum_{\mathbf{x}_U \in \mathcal{S}^u} \Pr_{\mathcal{M}}[Y \in S | \mathbf{x}_K, x_i, \mathbf{x}_U] \Pr_{\pi}[\mathbf{x}_U | \mathbf{x}_K, x_i] \\ &= \sum_{\mathbf{x}_U \in \mathcal{S}^u} \Pr[Y \in S | \mathbf{x}_K, x_i, \mathbf{x}_U] \Pr[\mathbf{x}_U | \mathbf{x}_K, x'_i] \frac{\Pr[\mathbf{x}_U | \mathbf{x}_K, x_i]}{\Pr[\mathbf{x}_U | \mathbf{x}_K, x'_i]} \\ &\stackrel{(\text{Lemma 7.4})}{\leq} \sum_{\mathbf{x}_U \in \mathcal{S}^u} \Pr[Y \in S | \mathbf{x}_K, x_i, \mathbf{x}_U] \Pr[\mathbf{x}_U | \mathbf{x}_K, x'_i] \left(\frac{\max_{kl} P_{kl}}{\min_{kl} P_{kl}} \right)^4 \\ &= \left(\frac{\max_{kl} P_{kl}}{\min_{kl} P_{kl}} \right)^4 \sum_{\mathbf{x}_U \in \mathcal{S}^u} \Pr[Y \in S | \mathbf{x}_K, x_i, \mathbf{x}_U] \Pr[\mathbf{x}_U | \mathbf{x}_K, x'_i] \\ &\leq \left(\frac{\max_{kl} P_{kl}}{\min_{kl} P_{kl}} \right)^4 e^{\varepsilon} \sum_{\mathbf{x}_U \in \mathcal{S}^u} \Pr[Y \in S | \mathbf{x}_K, x'_i, \mathbf{x}_U] \Pr[\mathbf{x}_U | \mathbf{x}_K, x'_i] \\ &= \left(\frac{\max_{kl} P_{kl}}{\min_{kl} P_{kl}} \right)^4 e^{\varepsilon} \Pr[Y \in S | \mathbf{x}_K, x'_i] \end{aligned}$$

Therefore, for every possible adversary with $u \geq 1$ we have that $\text{BDPL}_{(K,i)} \leq \varepsilon + 4 \ln \gamma$. Since we have bounded the $\text{BDPL}_{(K,i)}$ of every possible adversary (K, i) , we also bound the total BDPL. $\square$

(H1) means that all probabilities in P are greater than zero while (H2) imposes that initial distribution w must be an eigenvector with eigenvalue 1 of the transition matrix P , i.e., w is a *stationary distribution* i.e., the distribution of states does not change as the Markov chain progresses from time step to time step. Note that if (H1) holds, then for any irreducible and aperiodic Markov chain having s states, there exists at least one stationary distribution [10].

This theorem allows us to systematically create BDP mechanisms by recalculating the noise addition of existing DP mechanisms. The noise calibration only depends on the maximum ratio between the Markov transition probabilities of the model, γ , and the adjusted mechanism must be calibrated to $\varepsilon' = \varepsilon - 4 \ln(\gamma)$. Note that the best BDPL we can achieve is $\varepsilon = 4 \ln(\gamma)$ since $\varepsilon' \geq 0$.

For $n \in \mathbb{N}$, privacy parameter $\varepsilon > 0$, and maximum transition probability ratio $\gamma \geq 1$ we have

$$\varepsilon + 4 \ln \gamma < n\varepsilon \quad \text{if and only if} \quad \gamma < \exp\left(\frac{n-1}{4}\varepsilon\right). \quad (7.48)$$

Equation (7.48) and Figure 3 show that the Markov chain bound improves upon the general bound in most cases. For an ε -DP mechanism with $\varepsilon = 0.5$ and database of size $n = 80$, the Markov chain bound is an improvement even if the largest transition probability is 10 000 times the smallest. However, while we often see an improvement, this is not always the case. For instance, for $\varepsilon = 0.5$, $n = 20$ and a ratio γ of 100, the general bound would be smaller.

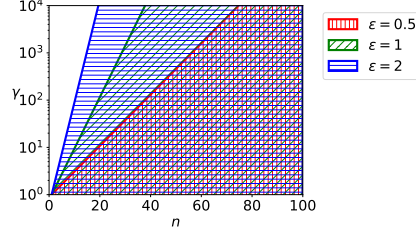


FIGURE 3. Comparison of Markov chain bound to general bound. The x-axis represents the size of the database, while the logarithmic y-axis shows the ratio γ . The Markov chain bound of the BDPL for an ε -DP mechanism improves upon the general bound for values of n and γ in the respective shaded area.

7.2. Accuracy. The Markov chain bound enables us to derive improved utility guarantees for the Laplace mechanism when γ is sufficiently small.

Corollary 7.6. *Let $\mathcal{M}_{\varepsilon,f}$ be the ε -Laplace mechanism that approximates the query $f : \mathcal{S}^n \rightarrow \mathbb{R}$ and inputs a database drawn from a Markov chain satisfying (H1) and (H2). If $\mathcal{M}_{\varepsilon,f}$ is (α, β) -accurate w.r.t. f and $\varepsilon \geq 4 \ln(\gamma)$ then, there exists an ε -BDP mechanism $\mathcal{B}$ that is $(h\alpha, \beta)$ -accurate w.r.t. f with*

$$h = \frac{\varepsilon}{\varepsilon - 4 \ln(\gamma)}.$$

Proof. The idea of this proof is to construct mechanism $\mathcal{B}$ with the Laplace mechanism as well, but to choose a carefully selected privacy leakage $\varepsilon' < \varepsilon$ so that mechanism $\mathcal{B}$ is (1) ε -BDP and (2) $(h\alpha, \beta)$ -accurate.

First, we determine the accuracy of mechanism $\mathcal{M}_{\varepsilon,f}$. With Proposition 3.5, we know that the (α, β) -accuracy of the Laplace mechanism for a given probability $\beta \in (0, 1]$ and privacy parameter ε is

$$\alpha = \ln\left(\frac{1}{\beta}\right) \frac{\Delta f}{\varepsilon}.$$

So this is the (α, β) -accuracy of $\mathcal{M}_{\varepsilon,f}$.

We have to show that there exists an ε -BDP mechanism $\mathcal{B}$ which is $(h\alpha, \beta)$ -accurate. We choose $\mathcal{B}$ as the Laplace mechanism applied to f with an adjusted privacy parameter $\varepsilon' > 0$. Thus, $\mathcal{B}$ will be ε' -DP. Therefore, we can use Theorem 7.5 to show that $\mathcal{B}$ is BDP. We must choose ε' in a way that ensures that the BDPL is limited to ε , so that we have ε -BDP. With Theorem 7.5, $\mathcal{B}$ is

$$(\varepsilon' + 4 \ln \gamma)\text{-BDP}.$$

Therefore, to achieve ε -BDP, we must have

$$\varepsilon' + 4 \ln \gamma = \varepsilon \Leftrightarrow \varepsilon' = \varepsilon - 4 \ln \gamma$$

Now, we can calculate the accuracy of $\mathcal{B}$ because it also uses the Laplace mechanism. Then, we find an upper bound for this accuracy. Mechanism $\mathcal{B}$ is (α', β) -accurate, with

$$\alpha' = \ln\left(\frac{1}{\beta}\right) \frac{\Delta f}{\varepsilon'} = \ln\left(\frac{1}{\beta}\right) \frac{\Delta f}{\varepsilon - 4 \ln \gamma} = \alpha h.$$

□

The statement of Corollary 7.6 is visualized in Figure 4. This figure shows that in order to provide similar utility guarantees to DP, either the BDPL bound ε has to be larger than 5, or the ratio γ between different transition probabilities must be smaller than 3.

In conclusion, the Markov-specific bound on the BDPL of any ε -DP mechanism improves upon the general bound under certain conditions, enabling us to provide new utility guarantees in Corollary 7.6 for mechanisms utilizing the Laplace mechanism. Notably, the positive impact of this bound becomes significant when the number of correlated records is large, as the general bound increases linearly with the number of correlated records, whereas the Markov bound

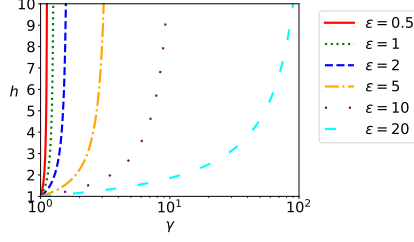


FIGURE 4. Relative accuracy of an ε -BDP to an ε -DP mechanism for Markov chain data. The logarithmic x-axis shows the transition probability ratio γ , while the y-axis shows the factor h . Given a Markov chain with transition probability γ , each line indicates the maximum error increase h .

remains unaffected by this parameter. However, it does have the limitation that the minimum privacy level that can be offered is constrained by the distribution properties, a limitation not present when applying the general bound.

8. UTILITY EXPERIMENTS

Theoretical bounds on privacy and utility do not always translate directly to practical implementations. For instance, while it may be theoretically feasible to achieve a given (α, β) -accuracy, designing or implementing a mechanism that attains this in practice can be challenging. In this section, we use our theoretical results to construct a BDP mechanism and empirically evaluate its utility on real-world databases modeled by either multivariate Gaussian correlations or Markov chains. Our objective is to demonstrate that the utility gains predicted under specific correlation structures, rather than arbitrary ones, are indeed achievable in practice.

Our theorems express utility in terms of (α, β) -accuracy. We use confidence intervals as the empirical counterpart for comparison between theoretical and practical estimates. If the upper bound of a $(1 - \beta)$ -confidence interval for the absolute error is close to the theoretical value α , this indicates that empirical results closely align with theoretical expectations, suggesting their tightness.

We aim to test the specific correlation models for which we derived new bounds, namely the Gaussian and Markov models. For this, we require one real database for each model, ensuring that the data follows the respective correlation structure and satisfies the additional conditions in our theorems. Specifically, correlation must be limited to $\rho < \frac{1}{n-2}$ for the Gaussian database, and for the Markov model, all transition probabilities must be positive, with an initial stationary distribution. We detail our database selection process to meet these criteria in Section 8.1.

We calibrated the Laplace mechanism using Theorem 6.9 and Theorem 7.5 to derive BDP mechanisms for each case. We then ran these BDP mechanisms on the selected databases and compared the utility results with those of BDP mechanisms designed to protect against arbitrary correlation, in order to assess the improvements offered by the correlation-specific approach. Additionally, we plotted the utility metric of a DP mechanism (i.e., the classical Laplace mechanism) as a baseline, representing the best-case utility achievable in the absence of correlation.

8.1. Databases. We use two real-world databases, one for each correlation model. The selection criteria are public availability and quality of the databases, as well as the fulfillment of the theoretical assumptions, namely, following the correlation model and fulfilling the extra hypotheses of the corresponding theorem in each case, regarding the Pearson correlation coefficient and the transition matrix.

8.1.1. Multivariate Gaussian: Galton Height Data. A historical database of heights of individuals [16] originally used to study the correlation between parents and children height. It is especially famous in statistical analysis because of its contribution to the understanding of correlation and linear regression. Particularly, it was used to define the concept of regression for the

first time [5]. This database is considered the classic representative of a linear correlation model, where the regression and correlation are interpreted in the context of a multivariate Gaussian distribution [28]. This makes it a suitable candidate for understanding the transferability of our Gaussian bound to practice.

The database comprises 897 records that contain the height of a person, their gender, an identifier for their family, both their parents heights, and the number of children in their family. The data is collected from 205 different families. In the context of our work, this translates to 205 sub-databases of size $n = 3$ correlated records. The empirical Pearson correlation coefficient is 0.275 between child and father, 0.202 between child and mother, and 0.074 between father and mother. Thus, we have a maximum Pearson correlation coefficient of $\rho = 0.275$ satisfying $\rho = 0.275 < 1 = \frac{1}{n-2}$. Therefore, this database fulfills all requirements to apply Theorem 6.9.

8.1.2. Markov Model: Activity Data. This data comprises the time series of the number of steps a single person performs during a day. It was recorded from a personal activity monitoring device collecting data at 5 minute intervals during the months of October and November 2012 [29]. We therefore test the event-level privacy of the stream from a local perspective, i.e., without assuming any trusted data curator [15].

If we order the activity data chronologically and capture the change of state, for instance from “walking”, to “running”, to “idle”, etc, we observe that the next state depends on the previous state. For the sake of interpretability we analyze the simplest Markov model on this data with two possible states: “inactive” when the user takes no steps in a 5 minutes interval, and “active” for intervals with more than 0 steps.

While this database is not as representative and widely used in research as the Galton height data, it nonetheless presents a realistic use case for data analysis involving a Markov chain model and private data. Our aim is to give an intuition for the transferability of our theoretical utility results to practice. To also measure the impact of database size on the utility of BDP mechanisms under Markov chain correlation, we prepare two versions of the database for our experiment: *Complete data*, where we combine the data of all days into a single database with $n = 17\,568$ records, and *Data for single days*, where we split the database into 61 unique subdatabases, each corresponding to the activity states of a single day with $n = 288$ records.

In order to fulfill the hypotheses of Theorem 7.5 and Corollary 7.6 we require the transition probabilities of the Markov chain to be positive. We calculate them empirically from the database containing all 17 568 records and receive the following transition matrix:

$$P = \begin{pmatrix} P_{00} & P_{10} \\ P_{01} & P_{11} \end{pmatrix} = \begin{pmatrix} 0.882 & 0.118 \\ 0.305 & 0.695 \end{pmatrix},$$

where $s = 0$ represents inactive state and $s = 1$ active state. Our theoretical results also require w to be a stationary distribution. While w can not be empirically computed since we only have one initial state, this Markov chain is irreducible, since both states are reachable from each other, aperiodic, since $P_{ss} \neq 0$ for both $s \in \{0, 1\}$, and $P_{st} > 0$ therefore there exists a stationary initial distribution [10]. Therefore, we conclude that the database fulfills the hypotheses for testing our results.

8.2. Target Queries and Utility metrics. We focus our utility study on two concrete although commonly used queries: sum and counting queries. Formally, given a database $D = (x_1, x_2, \dots, x_n)$, where each x_i represents a numerical value, a sum query is defined as:

$$q_S(D) = \sum_{i=1}^n x_i.$$

In the case of the Galton height data, each x_i corresponds to an individual’s height. If each record is binary, i.e., $x_i \in \{0, 1\}$, as is the case for the activity databases, $q_S(D)$ is called a counting query since it outputs the count of states with the attribute 1, in our case, the number of active states.

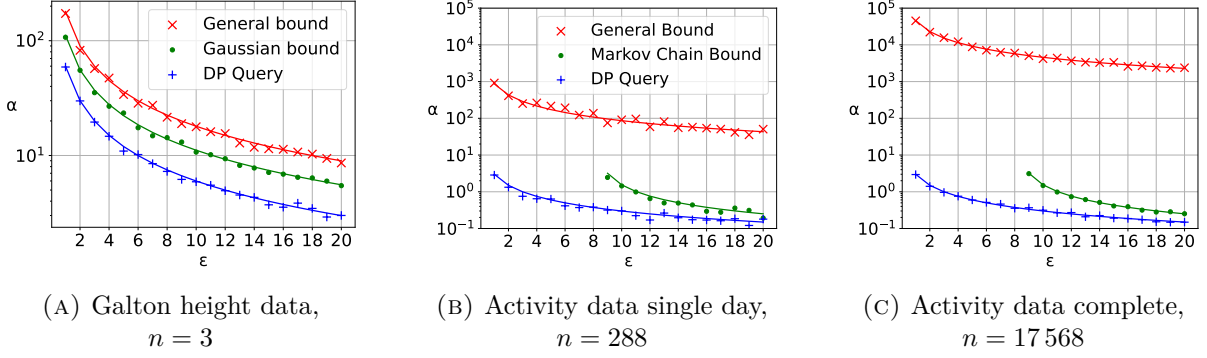


FIGURE 5. Error of private sum queries on Galton height data and private counting queries on activity data. The x-axis shows the privacy leakage ϵ of the BDP or DP query. The y-axis shows the absolute error α relative to the deterministic sum query on a logarithmic scale. The lines represent the expected theoretical result for error α with probability $\beta = 5\%$. The marks show the empirical upper limit of the 95% confidence interval of the absolute error.

In the Galton data, we compute the sum query for each subdatabase of correlated individuals, i.e., each family group. The real value (without any privacy) of the height sum query has an average output of 508cm for each family.

In the activity data for single days, the average count of active states per day is 80. In the complete activity data the count is 4250 occurrences of the active state.

We measure utility using the upper limit of a 95% confidence interval for the absolute query error of the private mechanisms. A smaller upper limit indicates higher utility, and vice versa. This utility metric is chosen due to its direct relation to (α, β) -accuracy: with $\beta = 0.05$ (corresponding to a 95% confidence level), we expect the empirical confidence interval limit to be close to the theoretical error α , validating our theoretical results. We select a 95% confidence interval due to its common use in practice [23]. To facilitate comparison with our theoretical results, we plot the theoretical expected error tolerance α for each mechanism, derived from Proposition 3.5 for the baseline DP mechanism and Corollary 5.4, Corollary 6.10, and Corollary 7.6 for the general bound, the Gaussian bound and the Markov chain bound respectively. Additionally, to give an intuition of the utility impact in practice, we report the mean absolute percentage error (MAPE) to estimate the expected relative error for a single execution. For a ground-truth function f , a set of m databases D_i , and a private mechanism $\mathcal{M}_f$, it is defined as

$$\text{MAPE} = \frac{1}{m} \sum_{i=1}^m \left| \frac{\mathcal{M}_f(D_i) - f(D_i)}{f(D_i)} \right|.$$

It is an estimation of the expected relative error for a single execution.

8.3. Mechanism and Experiment Design. In order to provide mechanisms able to guarantee BDP and approximate the target queries presented in Section 8.2, we use the Laplace mechanism with the noise calibrated through Theorem 5.2 for the baseline, Theorem 6.9 for Galton height data (Gaussian) and Theorem 7.5 for the activity database (Markov). In this section, we refer to the DP privacy leakage by τ , to avoid confusion with the maximum BDPL denoted by ϵ .

8.3.1. Height Data. As explained in Section 8.1, we assume that the height data is drawn from a multivariate Gaussian distribution with $\rho \leq 0.27$ for the Gaussian bound and sub-databases of correlated records of size $n = 3$. For $n = 3$, Theorem 6.9 tells us that the Laplace mechanism $\mathcal{M}_{\tau, f}$ applied to a sum query f is ϵ -BDP, with

$$\epsilon = \left(\frac{n^2}{4(\frac{1}{\rho} - n + 2)} + 1 \right) \tau \approx 1.853\tau. \quad (8.1)$$

In comparison, the general bound indicates that for the $\mathcal{M}_{\tau,f}$, we have $\varepsilon = m\tau = 3\tau$. We fix BDPL values $\varepsilon \in (0, 20]$ and compute the corresponding τ using Equation (8.1) for the Gaussian-specific mechanism and $\tau = \frac{\varepsilon}{3}$ for the general correlation approach. For $\varepsilon \in (0, 5)$, we ensure strong theoretical privacy guarantees, while also considering the higher range $\varepsilon \in [5, 20]$, which has shown empirical resilience to certain privacy attacks [6, 31].

All records are clipped to the range of 152.40cm to 203.2cm (60 to 80 inches).

The empirical errors are obtained by running the mechanisms on each subdatabase and comparing each result to the unperturbed non-private query result.

8.3.2. Activity Data. As discussed in Section 8.1 we assume that the data follows a Markov chain with probability ratio $\gamma \approx 7.47$. Theorem 7.5 tells us that the Laplace mechanism $\mathcal{M}_{\tau,f}$ applied to a counting query f is ε -BDP, with

$$\varepsilon = \tau + 4 \ln \frac{0.882}{0.118} = \tau + 8.05. \quad (8.2)$$

In comparison, with the general bound we have $\varepsilon = n\tau = 3\tau$ for mechanism $\mathcal{M}_{\tau,f}$. Analogous to the Gaussian case, we apply the Laplace mechanism to compute the sum query of each subgroup. Here, we fix BDPL values $\varepsilon \in (0, 20]$ and compute the corresponding τ using Equation (8.2) for the Markov-specific mechanism and taking $\tau = \frac{\varepsilon}{n}$ for $n = 228$ (daily databases) and $n = 17568$ (complete database) for the general correlation approach.

When we consider the situation of the complete database with $n = 17568$ records, we have only one database available—in contrast to the Galton height data split into 205 databases and the daily activity data split into 61 databases. Therefore, in order to be able to calculate empirical confidence intervals, we execute the mechanism for the complete activity data 1 000 times on the same database.

8.4. Results and Discussion. The results for the Galton height database, showing the Gaussian bound, the general bound, and a DP sum query, are presented in Figure 5A. We plot the DP mechanism as the baseline for the best possible utility; however, it is important to note that DP does not offer meaningful protection in this experiment, given the data’s correlations. Among the correlation-protecting mechanisms, those using the Gaussian bound consistently outperform those using the general bound for all ε values. Moreover, the empirical confidence intervals closely align with the theoretical error α , validating our theoretical results. Note that the Galton height data uses imperial units (inches), thus the errors are also interpreted in inches.

The results for the activity data are shown in Figure 5B and 5C. Again, we use the DP mechanism as the baseline for the best possible utility, not as a comparable protective mechanism. The DP query without correlation protection produces the smallest errors. For BDP mechanisms we observe that, for all the cases in which we can use the Markov chain bound ($\varepsilon \geq 9$), these mechanisms drastically outperform the ones based on the general bound. The benefit of using the Markov bound is especially pronounced in the complete activity data (Figure 5b), where the error of the mechanism using the general bound increases sharply while the mechanism with the Markov chain bound and the DP mechanism remain stable. This is because the general bound scales with the size of the database n , while the Markov bound is independent of n .

However, we must point out the limitation on the applicability of the Markov bound. While the general bound can provide an ε -BDP mechanism for any $\varepsilon > 0$, as explained in Section 7, the Markov bound only works for $\varepsilon \geq \tau - 4 \ln(\gamma)$, since τ -DP mechanisms require $\tau \geq 0$, corresponding to $\varepsilon \geq 9$ in our experimental data.

Finally, we see that the empirical results in Figure 5b match the theoretical error even closer than for Figure 5a. This may be explained by the larger sample size: For Figure 5a, each empirical confidence interval is calculated from 61 errors on 61 databases. For Figure 5b, there is only one database, but the mechanisms are executed 1 000 times, producing 1 000 measurements, resulting in the construction of more accurate confidence intervals.

The results demonstrate that BDP mechanisms calibrated with our newly proven Gaussian and Markov chain bounds outperform BDP mechanisms calibrated with the general bound in

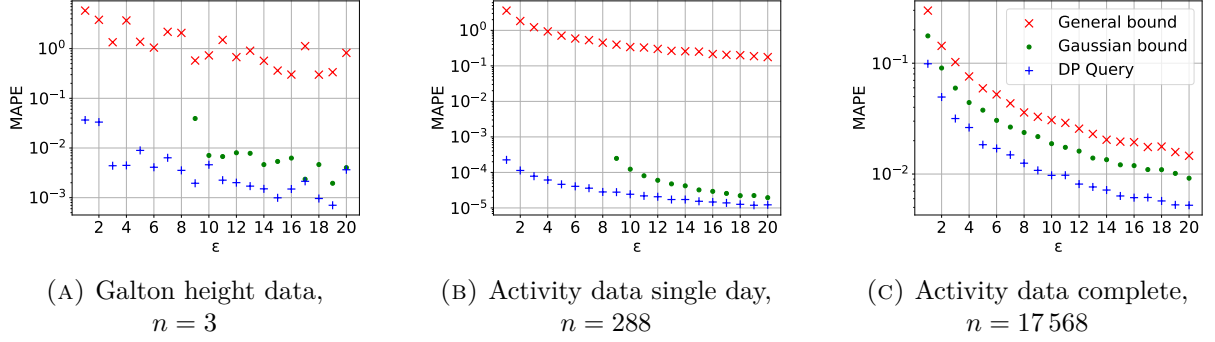


FIGURE 6. MAPE of private sum queries on Galton height data and private counting queries on activity data. The x-axis shows the privacy leakage ε of the BDP or DP query. The y-axis shows the MAE relative to the deterministic counting query on a logarithmic scale.

terms of utility on real-world data. Moreover, the empirical errors from our experiments closely align with our theoretical utility results, validating the practical applicability of our theorems.

We extend this study with the analysis of the relative error. Figure 6(B) and Figure 6(B)(C) show the MAPE for activity data (Markov chain model), Figure 6(A) for Galton height data (Gaussian correlation model). As expected, the DP query has the lowest MAPE however it does not offer protection against correlation. When we offer BDP protections we see that the correlation-specific BDP mechanisms (i.e., using the Gaussian bound or the Markov chain bound) outperform the BDP mechanism protecting against arbitrary correlation with the general bound. For activity data, the MAPE of the general bound reaches values above 100%, resulting in an error as large as the ground truth itself. In comparison, the Markov chain bound achieves errors below 10% for single day activity data, and below 0.1% for the complete activity data. The result for Galton height data is more favorable to the general bound; the worst-case error remains below 100% but above 10%. In this case, the relative scale of the Laplacian noise is smaller because the height values are clipped to the range 60 to 80 inches. Still, the Gaussian bound improves over the general bound and results in errors roughly half as large.

We acknowledge certain limitations when extrapolating our results. The validity of our experimental findings is constrained by the specific databases used. While the Galton height data serves as a well-known example of record correlation, it reflects only one of many possible correlation patterns. Similarly, most practical applications of a Markov chain would involve more than two states, introducing complexity beyond the binary-state model used in our study. Nevertheless, our results provide valuable insight into the practical applicability of our theorems and indicate their potential for real-world scenarios. Furthermore, these experiments demonstrate that achieving meaningful utility while protecting against correlation is feasible in practice.

9. CONCLUSION

In this paper we have investigated the utility of BDP for correlated data. We address the previous literature limitations by expanding upon new correlation models and providing a thorough analysis of privacy-utility trade-offs, grounded in both theoretical insights and empirical validation. Explicitly, we provide new relationships between DP and BDP mechanisms and how these can be used for privacy protection in the presence of correlations.

We are the first to prove that ε -DP implies $m\varepsilon$ -BDP, where m is the maximum size of a set of correlated records. This result is tight even when the strength of correlation is arbitrarily small. We improve on this bound by restricting the correlation model to a multivariate Gaussian distribution and a Markov chain, obtaining novel bounds on the BDP leakage that improve over previous assumptions and yield much better utility with identical data protection.

We conclude that it is possible to achieve better utility without weakening the adversary model, at least in the following situations: (a) when the size of the sets of correlated records is

sufficiently small (depending on the specific task and error tolerance); (b) when the data follows a multivariate Gaussian correlation model with weak correlation ($\rho \approx \frac{1}{n}$); or (c) when the data is a time series following a Markov chain with sufficiently similar transition probabilities.

We also conclude that the application of BDP without assuming the data correlation model is a futile attempt. Precisely, the utility of any ε -BDP mechanism must be poor if it protects against arbitrary correlations, particularly when all records may be correlated and statistically significant privacy guarantees are required ($\varepsilon \in (0, 5)$).

Furthermore, our experiments testing the utility of BDP mechanisms on real-world data confirm that assuming a specific correlation model can significantly improve utility compared to the general bound. Additionally, we observed that the utility of mechanisms based on the general bound is highly sensitive to the number of correlated records, highlighting the advantage of using our Markov bound, which is agnostic to the data size, when we have large sets of correlated records.

Overall, our Theorems 5.2, 6.9 and 7.5 advance the understanding of the relationship between the privacy concepts of DP and BDP, facilitating the use of existing DP mechanisms as BDP mechanisms. This opens up future research avenues where correlation-specific bounds can be derived to create BDP mechanisms with enhanced utility, guiding practitioners in determining when limiting BDP leakage is a viable strategy for protecting against privacy attacks that exploit correlations between records.

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